MATHEMATICAL STATISTICS

Contents

1. Basic Setup

Definition 1.1 (Statistical setup)**.** A *statistical setup* consists of the following data:

We have the following measurable spaces:

- The *observation space* $\mathcal{X}$.

- The *parameter space* Θ.
- The *action space* $\mathcal{A}$.
- The *sample space* Ω

We also have the following maps:

- A *loss function* $L : \Theta \times \mathcal{A} \to \mathbb{R}$.
- An *observation* $X : \Omega \to \mathcal{X}$.
- A *decision function* $d : \mathcal{X} \to \mathcal{A}$.

For each parameter $\theta \in \Theta$, we have a probability measure $P_\theta \in \mathrm{Prob}(\Omega)$. The associated risk function is then

$$\begin{aligned} R(\theta, d) &= E_\theta L(\theta, d(X)) \\ &= \int_\Omega L(\theta, d(X(\omega)) dP_\theta(\omega). \end{aligned}$$

Note: It seems that unless otherwise specified, it does not hurt to just assume $\mathcal{X} = \Omega$ and X is the identity map in which case

$$R(\theta, d) = \int_{\mathcal{X}} L(\theta, d(x)) dP_\theta(x).$$

Interpretation: We have an unknown parameter θ and a given observation $X(\omega) \in \mathcal{X}$ where $\omega \in \Omega$ was randomly sampled according to P_θ. Based on this observation, we take an action $d(x) \in \mathcal{A}$. Then the loss $L(\theta, d(x))$ measures how wrong we are in taking this action $d(x)$. The risk $R(\theta, d)$ is then the average of this loss (for the fixed θ).

Example 1.2. Suppose now $\Omega = \mathcal{X} = \mathbb{R}^n$ and X is identity (as usual) and $\mathcal{A} = \mathbb{R}$. Let $\Theta = \mathbb{R} \times \mathbb{R}_{\geq 0}$ and for $\theta = (\mu, \sigma^2) \in \Omega$ we let P_θ denote the distribution of the random variable $(X_1, \ldots, X_n) \in \mathbb{R}^n$ where the X_i are i.i.d $N(\mu, \sigma^2)$. We take a loss function $L(\theta, a) = |\mu - a|^2$. Thus if $d : X \to \mathcal{A}$ is a decision rule then $L(\theta, d(x))$ will measure how well of an approximation $d(x) \in \mathbb{R}$ is of the unknown μ. Let us choose $d(x_1, \ldots, x_n) = \frac{1}{n}\sum_{i=1}^n x_i$. Thus

$$R(\theta, d) = E|\frac{1}{n}\sum_{i=1}^N X_i - \mu|^2 = Var(\frac{1}{n}(X_1 + \ldots + X_n)) = \frac{\sigma^2}{n}$$

by the independence of the X_i. We see that as $R(\theta, d) = O(n^{-1})$ so this seems to be a good decision rule.

If instead we use the loss function $L(\theta, a) = |\sigma^2 - a|$ then a good decision rule seems to be

$$d(X) = s^2 := \frac{1}{n-1}\sum_{i=1}^n |X_i - \overline{X}|^2.$$

It turns out that (TODO?)

$$R(\theta, d) = \frac{2\sigma^4}{n-1}.$$

We can use this language to define hypothesis testing:

Definition 1.3. Suppose now that we have a partition $\Theta = \Theta_0 \sqcup \Theta_1$ of the parameter space and our action space is $\mathcal{A} = \{\Theta_0, \Theta_1\}$. We have a $0-1$ loss function given by

$$L(\theta, \Theta_i) = 1 - \mathbb{1}_{\Theta_i}(\theta).$$

Thus $L(\theta, a)$ measures whether we chose the right partition $a \in \mathcal{A}$ that θ landed in. We see that the risk

$$R(\theta, d) = P_\theta(\{\omega \in \Omega \mid \theta \notin d(X(\omega))\})$$

is the probability that our decision algorithm chooses the wrong partition.

The action Θ_0 is called the *Null Hypothesis* while Θ_1 is called the *Alternative hypothesis.* A *Type I* error is an action that chooses the alternative hypothesis (Θ_1) when the null hypotheiss is true ($\theta \in \Theta_0$). A *Type II error* is an action that chooses the null hypothesis when the alternative is true.

Example 1.4. Suppose we have a dice that is either fair (null hypothesis) or always rolls a 6 (alternative hypothesis). Suppose we have rolled this dice n times and observed the outcomes $(X_1, \dots, X_n) \in \{1, 2, 3, 4, 5, 6\}^n = \mathcal{X}$. Say $\Theta = \{\theta_0, \theta_1\}$ where P_{θ_0} is the uniform distribution on $\mathcal{X}$ (dice is fair) while P_{θ_1} is the probability measure concentrated on $(6, \dots, 6) \in \mathcal{X}$. Our decision rule simply chooses θ_1 if and only if we roll all 6. We have $R(\theta_0, d) = 1/6^n$ while $R(\theta_1, d) = 0$.

Definition 1.5. We say that a statistical setup is *nice* if there exists a σ-finite positive measure (not necessarily a probability measure) ν on $\Omega = \mathcal{X}$ and a measurable function $f : \mathcal{X} \times \Theta \to \mathbb{R}_{\geq 0}$ such that $P_\theta \ll \nu$ and

$$\frac{dP_\theta}{d\nu}(x) = f(x|\theta) := f(x, \theta)$$

for all $\theta \in \Theta$.

For example, if $\mathcal{X}$ is countable and thus each P_θ is discrete, then ν can be the counting measure.

Assumption going forward: We always assume a statistical setup is nice without mentioning it. This is the case for the examples above (i.e., ν is either a Lebesgue measure or a counting measure).

2. Bayesian Setup

Definition 2.1. A *prior distribution* is a probability measure μ_Θ on the parameter space Θ.

Recalling that we assume the nice condition of Definition 1.5, we thus have a probability measure P on $\mathcal{X} \times \Theta$ given by

$$\frac{dP}{d(\nu \times \mu_\Theta)}(x, \theta) = f(x|\theta).$$

In other words for integrable $\psi : \mathcal{X} \times \Theta \to \mathbb{R}$ we have

$$\begin{aligned}
\int_{\mathcal{X}\times\Theta} \psi(x, \theta) dP(x, \theta) &= \int_{\mathcal{X}\times\Theta} \psi(x, \theta) f(x|\theta) d(\nu \times \mu_\Theta)(x, \theta) \\
&= \int_\Theta \left(\int_{\mathcal{X}} \psi(x, \theta) f(x|\theta) d\nu(x) \right) d\mu_\Theta(\theta) \\
&= \int_\Theta \int_{\mathcal{X}} \psi(x, \theta) dP_\theta(x) d\mu_\Theta(\theta)
\end{aligned}$$

From now on, we let $X : \mathcal{X} \times \Theta \to \mathcal{X}$ and $\Theta : \mathcal{X} \times \Theta \to \Theta$ denote the projections (note: by abuse of notation Θ is now either a space or a projection map, but it should cause no confusion in any context).

Let $P_X = X^* P$ denote the push-forward measure of P onto $\mathcal{X}$, that is, $P_X(A) = P(A \times \Theta)$ for $A \subset \mathcal{X}$. We let

$$g(x) := \int_\Theta f(x, \theta) d\mu_\Theta(\theta).$$

Observe that for $A \subset \mathcal{X}$ we have that

$$P_X(A) = \int_{A\times\Theta} f(x|\theta)d\mu_\Theta(\theta)d\nu(x) = \int_A g(x)d\nu(x)$$

and thus P_X has a density with respect to ν given by g.

Lemma 2.2. For P_X-almost all $x \in \mathcal{X}$ we have that

$$g(x) > 0.$$

Proof. Let $A = \{x \in \mathcal{X} \mid g(x) = 0\}$. Then

$$P_X(A) = \int_A g(x)d\nu(x) = 0.$$

□

Given this lemma, we can now define for P_X-almost all $x \in \mathcal{X}$ a probability measure on Θ given by

$$P_{\Theta|X=x}(B) = \frac{1}{g(x)} \int_B f(x|\theta)d\mu_\Theta(\theta)$$

for $B \subset \Theta$. This is known as the *posterior distribution of* Θ *given* $X = x$.

Proposition 2.3. The measures $\delta_x \times P_{\Theta|X=x}$ are conditional distributions of P given $X = x$ where $X : \mathcal{X} \times \Theta \to \mathcal{X}$ is the projection. The conditional expectation of an integrable $\psi : \mathcal{X} \times \Theta \to \mathbb{R}$ given $X = x$ is given by

$$E(\psi|X = x) = \frac{1}{g(x)} \int_\Theta \psi(x,\theta)f(x|\theta)d\mu_\Theta(\theta).$$

Proof. Let $E'(\psi|X = x)$ denote the right hand side. We first show that it is well defined for almost all $x \in X$. This follows since by assumption we have that $|\psi(x,\theta)|$ is integrable with respect to P and thus

$$\int |\psi(x,\theta)|f(x|\theta)d\nu(x)d\mu_\Theta(\theta) < \infty.$$

It follows from Fubini-Tonelli that for ν almost all $x \in X$ we have that

$$\int_{\mathcal{X}} |\psi(x,\theta)|f(x|\theta)d\mu_\Theta(\theta) < \infty$$

and thus $E'(\psi|X = x)$ is a well defined measurable function on X. We have to show that $(x,\theta) \mapsto E'(\psi|X = x)$ is measurable with respect to the σ-algebra induced by $X : \mathcal{X} \times \Theta \to \mathcal{X}$ and that for all $A \subset X$ we have that

$$\int_{A\times\Theta} E'(\psi|X = x)dP(x,\theta) = \int_{A\times\Theta} \psi(x,\theta)dP(x,\theta).$$

The measurability is obvious as there is no dependence on Θ. For the second claim, note that since (as shown above) $g = dP_X/d\nu$, we have

$$\begin{aligned}
\int_{A\times\Theta} E'(\psi|X = x)dP(x,\theta) &= \int_A E'(\psi|X = x)dP_X(x) \\
&= \int_A E'(\psi|X = x)g(x)d\nu(x) \\
&= \int_A \int_\Theta \psi(x,\theta)f(x|\theta)d\mu_\Theta d\nu(x)
\end{aligned}$$

which completes the proof of this property since $f(x|\theta) = \frac{dP}{d(\nu\times\mu_\Theta)}(x,\theta)$. Finally, observe that

$$\int_{\mathcal{X}\times\Theta} \psi\ d(\delta_x \times P_{\Theta|X=x}) = \int_\Theta \psi(x,\theta)dP_{\Theta|X=x}(\theta) = \frac{1}{g(x)} \int_\Theta f(x|\theta)\psi(x,\theta)d\mu_\Theta(\theta) = E(\psi \mid X = x)$$

so indeed $\delta_x \times P_{\Theta|X=x}$ is the conditional probability of P given $X = x$. $\square$

Proposition 2.4. For $A \subset \mathcal{X}$ we have that

$$P(A \times \Theta \mid \Theta = \theta) = P_\theta(A).$$

Proof. By symmetry, if we swap the roles of $\mathcal{X}$ with Θ and apply the proposition above, then we get

$$\begin{aligned} P(A \times \Theta \mid \Theta = \theta) &= \left(\int_{\mathcal{X}} f(x|\theta) d\nu(x)\right)^{-1} \left(\int_A f(x|\theta) d\nu(x)\right) \\ &= P_\theta(X)^{-1} P_\theta(A) \\ &= P_\theta(A). \end{aligned}$$

$\square$

2.1. **Notation.** We use the standard notation from probability theory: if $F_i : \Omega \to \Omega_i$ are measurable and P is a measure on Ω, then

$$P(F_1 \in A_1, \ldots, F_n \in A_n) := P(\{\omega \in \Omega \mid F_1(\omega) \in A_1, \ldots, F_n(\omega) \in A_n\})$$

and likewise for conditional measures.

Thus if P is the measure on $\mathcal{X} \times \Omega$ as constructed above, then

$$P(X \in A, \Theta \in B) = P(A \times B).$$

3. Sufficient statistics

3.1. **Basic definitions and examples.** A statistic is a measurable function $T : \mathcal{X} \to \mathcal{T}$ to some measurable space $(\mathcal{T}, \mathcal{B}_{\mathcal{T}})$.

Definition 3.1. We say that a statistic T on $\mathcal{X}$ is *sufficient (in the Bayesian sense)* if for any prior μ_Θ on Θ we have for any Borel $B \subset \Theta$ we have the equality

$$P(\Theta \in B \mid X = x) = P(\Theta \in B \mid T \circ X = Tx) \quad \text{for } \mu_X\text{-almost all } x \in \mathcal{X}.$$

Example 3.2. Consider the case where $\mathcal{X} = \{0,1\}^n$ and $\Theta = [0,1]$. Suppose that P_θ is the Binomial probability distribution on $\{0,1\}^n$ where $(X_1, \ldots, X_n) \in \mathcal{X}$ is drawn i.i.d with $P(X_i = 1) = \theta$. Let us for now assume the Prior distribution μ_Θ is unknown. We know that

$$P_\theta(X = x) = \theta^{T(x)}(1-\theta)^{n-T(x)}$$

where $T(x) = \sum_{i=1}^n x_i$ is the number of successes. The posterior distribution is thus given by

$$P(\Theta \in B \mid X = x) = \frac{P(\Theta \in B, X = x)}{P(X = x)} = \frac{\int_B \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)}{\int_\Theta \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)}$$

for all $x \in \mathcal{X}$ that occur with positive probability (this is all of $\mathcal{X}$ except for possibly $x = (0, \ldots, 0)$ or $x = (1, \ldots, 1)$ in case θ is concentrated on 1 or 0 respectively).

We claim that $T : \mathcal{X} \to \mathbb{Z}_{\geq 0}$ given by $T(x) = \sum_{i=1}^n x_i$ is a sufficient statistic. Indeed if we fix $x_0 \in \mathcal{X} = \{0,1\}^n$ and we let $m = Tx_0$, we get that

$$\begin{aligned} P(\Theta \in B \mid T \circ X = m) &= \frac{P(\Theta \in B, T \circ X = m)}{P(T \circ X = m)} \\ &= \frac{\sum_{Tx=m} P(\Theta \in B, X = x)}{\sum_{Tx=m} P(X = x)} \\ &= \frac{\sum_{Tx=m} \int_B \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)}{\sum_{Tx=m} \int_\Omega \theta^{T(x)}(1-\theta)^{n-T(x)} d\mu_\Theta(\theta)} \\ &= \frac{\binom{n}{m} \int_B \theta^m (1-\theta)^{n-m} d\mu_\Theta(\theta)}{\binom{n}{m} \int_\Omega \theta^m (1-\theta)^{n-m} d\mu_\Theta(\theta)} \end{aligned}$$

And so we see that this fraction cancels out to the expression for $P(\Theta \in B \mid X = x_0)$ that we already obtained.

In fact, the same argument shows that T is still a sufficient statistic if instead of assuming that P_θ is binomial we assume that P_θ is *exchangable*, meaning a distribution on $\mathcal{X} = \{0,1\}^n$ that is invariant under permutations of the co-ordinates, in this case the same calculation will instead yield

$$P(\Theta \in B \mid TX = m) = \frac{\sum_{Tx=m} \int_B P_\theta(x) d\mu_\Theta}{\sum_{Tx=m} \int_\Omega P_\theta(x) d\mu_\Theta}$$

but since we assume that $P_\Theta(x) = P_\Theta(x')$ whenever x and x' are permutations of each other, we must have that all the summands are the same and hence we get the same cancellation.

Remark 3.3. Intuitively, sufficiency of a statistic essentially means that if we want to estimate the Posterior distribution of θ given an observable statistic, we only need to look at the statistic of $T \circ X$ rather than X, which may be simpler. In other words, somebody who wishes to estimate θ (e.g., find a confidence interval) need not know the sample x but just needs to know Tx. In this example, the person wishing to estimate θ needs only to be told a single integer Tx, rather than a potentially very long sequence $x \in \{0,1\}^n$ (data compression).

The classical definition of sufficiency does not make reference to a prior distribution and is thus not Bayesian.

Definition 3.4. We say that a Borel $A \subset \mathcal{X}$ is Θ-co-null if $P_\theta(A) = 1$ for all $\theta \in \Theta$. Likewise, if $T : \mathcal{X} \to \mathcal{T}$ is a statistic, then we say that $B \subset \mathcal{T}$ is Θ-co-null if $T^{-1}B$ is Θ-co-null, i.e., $(T^* P_\theta)(B) = 1$ for all $\theta \in \Theta$ (confusion may arise if we have another statistic $T' : \mathcal{X} \to \mathcal{T}$, but this tends to not be the case as we always deal with a single statistic at a time).

Definition 3.5. We say that a statistic $T : \mathcal{X} \to \mathcal{T}$ is *sufficient in the classical sense* if we have a Θ-co-null Borel set $\mathcal{T}_0 \subset \mathcal{T}$ and a family of measures ν_t, for $t \in \mathcal{T}_0$, such that for all $\theta \in \Theta$ and for Borel $B \subset \mathcal{X}$ we have that

$$P_\theta(B|T = t) = \nu_t(B) \quad \text{for } T^* P_\theta - \text{almost all } t \in \mathcal{T}$$

and $t \mapsto \nu_t(B)$ is Borel on $\mathcal{T}_0$.

This is a technical way of saying that T is sufficient if and only if $P_\theta(B|T = t)$ does not depend on θ. We stated the definition formally in order to deal with the issue that conditional probability is only almost everywhere unique and well defined.

Example 3.6. Let us continue from the binomial distribution in Example 3.2. As $\mathcal{X} = \{0,1\}^n$ is a finite set, we only have to consider the case where $B \subset \mathcal{X}$ is a singleton $B = \{x_0\}$ for some $x_0 \in \mathcal{X}$. Now $P_\theta(B|T = t) = 0$ if $Tx_0 \neq t$. If $Tx_0 = t$ then we have that

$$P_\theta(B|T = t) = \frac{P_\theta\{x_0\}}{P_\theta\{x \in \mathcal{X} \mid Tx = x_0\}} = \frac{1}{\binom{n}{t}}$$

where the last equality follows from the fact that all $x \in \mathcal{X}$ with $Tx = t$ have the same probability in a binomial distribution. In fact, the same argument applies to the case where the P_θ are all exchangable.

Thus we may take

$$\nu_t(\{x_0\}) = \mathbb{1}(Tx_0 = t)\binom{n}{t}^{-1}.$$

Thus this T is indeed sufficient in the classical sense.

3.2. **Equivalence of definitions.** In this section we prove equivalence of the classical and Bayesian definition of a sufficient statistics under the nice assumptions of Definition 1.5 (which we always assume).

Lemma 3.7. Suppose $T : \mathcal{X} \to \mathcal{T}$ is a statistic and μ_Θ is a prior on Θ. Fix Borel $A \subset \mathcal{X}$ and suppose that there is a Borel map $s : \mathcal{T} \times \Theta \to [0,1]$ so that

$$s(t,\theta) = P_\theta(A \mid T = t) \quad \text{for } T^*P_\theta\text{-almost all } t \in \mathcal{T}.$$

We have that

$$P(X \in A \mid TX = Tx, \Theta = \theta) = s(Tx,\theta) \quad \text{for } P\text{-almost all } (x,\theta) \in \mathcal{X} \times \Theta.$$

Proof. We must show that for $B \subset \Theta, C \subset \mathcal{T}$ we have that

$$\int_{T^{-1}C \times B} P(A \times \Theta \mid Tx = t, \Theta = \theta) dP(x,\theta) = \int_{T^{-1}C \times B} s(Tx,\theta) dP(x,\theta).$$

The law of total probability tells us that the left hand side is equal to

$$P((A \times \Theta) \cap (T^{-1}C \times B)) = P(X \in A \cap T^{-1}C, \Theta \in B).$$

The right hand side is

$$\begin{aligned}
\int_B \left(\int_{T^{-1}C} P_\theta(A|T = Tx) dP_\theta(x) \right) d\mu_\Theta(\theta) &= \int_B P_\theta(A \cap T^{-1}C) d\mu_\Theta(\theta) \\
&= \int_B \int_{\mathcal{X}} \mathbb{1}_{A \cap T^{-1}C}(x) dP_\theta(x) d\mu_\Theta(\theta) \\
&= P(X \in A \cap T^{-1}C, \Theta \in B)
\end{aligned}$$

□

Theorem 3.8. Suppose that $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the classical sense. Then T is also sufficient in the Bayesian sense (for any prior μ_Θ on Θ). Moreover for $A \subset \mathcal{X}$ we have

$$P(X \in A \mid TX = Tx, \Theta = \theta) = P(X \in A \mid TX = Tx) \quad \text{for } P\text{-almost all } (x,\theta) \in \mathcal{X} \times \Theta. \tag{1}$$

Proof. We first show the *Moreover* claim. By the definition of sufficiency we have that

$$P_\theta(A \mid T = t) = \nu_t(A)$$

for some measures $\nu_t, t \in \mathcal{T}_0$ where $\mathcal{T}_0 \subset \mathcal{X}$ is Θ-co-null. So indeed the Borel condition of the previous Lemma is satisfied with[1] $s(t,\theta) = \nu_t(A)$, where we see there is no dependence on θ. By the previous Lemma the left hand side of (1) is equal to $s(Tx,\theta)$. So now it remains to show that $s(Tx,\theta) = P(A \times \Theta \mid TX = Tx)$ for P-almost all (x,θ). Thus we must show that for $C \subset \mathcal{T}$ we have that

$$\int_{T^{-1}C\times\Theta} s(Tx,\theta)dP(x,\theta) = \int_{T^{-1}C\times\Theta} P(A\times\Theta \mid TX = Tx)dP(x,\theta). \tag{2}$$

By definition of conditional probabiltity w.r.t $TX : \mathcal{X}\times\Theta \to \mathcal{T}$, the right hand side is equal to

$$P(A\times\Theta \cap T^{-1}C\times\Theta) = P(X \in A\cap T^{-1}C).$$

Repeating the calculation of the previous Lemma we have that the left hand side of (2) is

$$\begin{aligned}\int_\Theta \left(\int_{T^{-1}C} P_\theta(A|T = Tx)dP_\theta(x)\right) d\mu_\Theta(\theta) &= \int_\Theta P_\theta(A\cap T^{-1}C)d\mu_\Theta(\theta)\\ &= \int_\Theta\int_{\mathcal{X}} \mathbb{1}_{A\cap T^{-1}C}(x)dP_\theta(x)d\mu_\Theta(\theta)\\ &= P(X\in A\cap T^{-1}C).\end{aligned}$$

This completes the proof of (2). Now it remains to show that T is sufficient in the Bayesian sense. By this just proved Moreover claim, it follows from Lemma D.4 that X and Θ are conditionally independent given TX. Using Lemma D.4 again we have that

$$P(\Theta\in B \mid TX = Tx, X = x) = P(\Theta \in B \mid TX = Tx) \qquad \text{for } P\text{-almost all } (x,\theta)$$

and thus for P_X-almost all $x \in \mathcal{X}$. But the left hand side is clearly $P(\Theta\in B \mid X = x)$. Thus T is indeed sufficient in the Bayesian sense. □

Lemma 3.9. A statistic $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the Bayesian sense if and only if for all prior distributions μ_Θ on Θ and measurable $B \subset \mathcal{X}$ we have that

$$x \mapsto P(\Theta\in B \mid x \in X)$$

is almost surely $T^{-1}\mathcal{B}_T$-measurable (i.e., almost surely equal to a function of T).

Proof. We prove the non-obvious direction only: Thus suppose that for each measurable $B\subset X$ we have that $x\mapsto P(\Theta\in B \mid x\in X) = h_B(Tx)$ where $h_B : \mathcal{T}\to\mathbb{R}$ is Borel. We first use apply the law of total probability to the map $T\circ X : \mathcal{X}\times\Theta\to\mathcal{T}$ as follows. For $A\subset\mathcal{T}$ Borel we have that

$$P(\Theta\in B, T\circ X\in A) = \int_A P(\Theta\in B|T\circ X = t)dP_{\mathcal{T}}(t).$$

Now we apply the law of total probability to the map $X : \mathcal{X}\times\Theta\to\mathcal{X}$ to obtain that

[1]We can just extend $s(t,\theta)$ arbitrarily to all of $\mathcal{T}$ to something Borel measureable (can set $s(t,\theta) = 0$ for $t \notin \mathcal{T}_0$) and the condition of the previous Lemma still holds.

$$\begin{aligned}
P(\Theta \in B, T \circ X \in A) &= P(\Theta \in B, X \in T^{-1}A) \\
&= \int_{T^{-1}A} P(\Theta \in B | X = x) dP_X(x) \\
&= \int_{T^{-1}A} h_B(Tx) dP_X(x) \\
&= \int_A h_B(t) dP_{\mathcal{T}}(t).
\end{aligned}$$

Thus we must have that $P(\Theta \in B | T \circ X = t) = h_B(t)$ holds for almost all t. Thus

$$P(\Theta \in B \mid T \circ X = Tx) = h_B(Tx) = P(\Theta \in B \mid X = x)$$

holds for almost all $x \in X$. □

Lemma 3.10. If $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the Bayesian sense then there exists a probability measure ν^* on $\mathcal{X}$ such that $P_\theta \ll \nu^*$ for all $\theta \in \Theta$, $\nu^* \ll \nu$ and for each $\theta \in \Theta$ there is a Borel

$$h_\theta : \mathcal{T} \to [0, \infty)$$

such that

$$\frac{dP_\theta}{d\nu^*}(x) = h_\theta(Tx) \quad \text{for } \nu^*\text{-almost all } x.$$

Proof. By Lemma E.2 there exists a countable or finite sequence of distinct $\theta_1, \theta_2, \ldots$ and $0 < c_1, c_2, \ldots \leq 1$ with $\sum_i c_i = 1$ such that $\nu^* := \sum_{i>0} c_i P_{\theta_i}$ satisfies that $P_\theta \ll \nu^* \ll \nu$ for all $\theta \in \Theta$. Now fix any $\theta_0 \in \Theta$. First suppose $\theta_0 \neq \theta_i$ for all $i > 0$. Define now a prior μ on Θ where $\mu(\{\theta_0\}) = 1/2$ and $\mu(\{\theta_i\}) = c_i/2$, i.e., μ is discrete. Now we have

$$\begin{aligned}
P(\Theta = \theta_0 \mid X = x) &= \frac{\int_{\{\theta_0\}} f(x|\theta) d\mu(\theta)}{\int_\Theta f(x|\theta) d\mu(\theta)} \\
&= \frac{\frac{1}{2} f(x|\theta_0)}{\frac{1}{2} f(x|\theta_0) + \frac{1}{2} \sum_{i>0} c_i f(x|\theta_i)}.
\end{aligned}$$

Observe that the pushfoward measure on $\mathcal{X}$ is $P_X = \frac{1}{2} P_{\theta_0} + \frac{1}{2} \nu^*$. We know by Lemma 2.2 that for P_X-almost all $x \in X$ the denominator is non-zero. Now observe that

$$\frac{d\nu^*}{d\nu}(x) = \sum_{i>0} c_i f(x|\theta_i)$$

and as $\nu^* \ll \nu$ we have that this is ν^* almost surely non-zero and thus P_X-almost surely non-zero as $P_X \ll \nu^*$ as shown above. So we can divide by this quantity to get that

$$\frac{dP_{\theta_0}}{d\nu^*(x)} = f(x|\theta_0) \left(\sum_{i>0} c_i f(x|\theta_i) \right)^{-1} \quad \text{for } P_X\text{-almost all } x \in \mathcal{X}. \tag{3}$$

Thus we have that

$$P(\Theta = \theta_0 \mid X = x) = \frac{dP_\theta / d\nu^*(x)}{dP_\theta / d\nu^*(x) + 1}$$

holds P_X-almost surely. However, P_X-almost surely we have that the right hand side is $\mathcal{T}$-measurable. Thus ν^*-almost surely it also is. This comlpetes the proof in the case $\theta_0 \neq \theta_i$ for all $i > 0$. If $\theta_0 = \theta_i$ for some $i > 0$ we instead define the prior

$$\mu = \sum_{i>0} c_i P_{\theta_i}.$$

Without loss of generality, say $\theta_0 = \theta_1$. Thus

$$P(\Theta = \theta_1 \mid X = x) = \frac{c_1 f(x|\theta_1)}{\sum_{i>0} c_i f(x|\theta_i)} = c_1 \frac{dP_{\theta_1}}{d\nu^*}(x).$$

Thus again we have that this is P_X-almost equal to a $\mathcal{T}$-measurable function (and thus ν^* almost equal to such).

□

Theorem 3.11. Assume the nice condition of Definition 1.5 . If $T : \mathcal{X} \to \mathcal{T}$ is sufficient in the Bayesian sense, then it is also sufficient in the classical sense. Letting ν^* be the measure from the previous Lemma, we have the more precise formulation of classical sufficiency: There is a $\mathcal{T}$-measurable set $\mathcal{X}' \subset \mathcal{X}$ such that for all $\theta \in \Theta$ we have $\nu^*(\mathcal{X}') = 1 = P_\theta(\mathcal{X}')$ and for $A \subset X$ we have that

$$P_\theta(A \mid T = Tx) = \nu^*(A \mid T = Tx) \quad \text{for all } x \in \mathcal{X}'.$$

Proof. Let ν^* be the measure from the previous Lemma and recall that

$$\frac{dP_\theta}{d\nu^*}(x) = h_\theta(Tx).$$

We will now show that for all $\theta \in \Theta$ and $A \subset X$ we have that

$$P_\theta(A \mid T = t) = \nu^*(A \mid T = t),$$

which will complete the proof. First, notice that $\nu^*(A \mid T = Tx)$ is defined on a $\mathcal{T}$-measurable subset $\mathcal{X}' \subset \mathcal{X}$ with $\nu^*(\mathcal{X}') = 1$. But as $P_\theta \ll \nu^*$ we have that that $P_\theta(\mathcal{X}') = 1$, which is a requirement for conditional probability. Now observe that for $\mathcal{T}$-measurable $C \subset \mathcal{X}'$ we have that

$$\begin{aligned}
\int_C \nu^*(A \mid T = Tx) dP_\theta(x) &= \int_C \nu^*(A \mid T = Tx) h_\theta(Tx) d\nu^*(x) \\
&= \int_{\mathcal{X}} \mathbb{E}(\mathbb{1}_A \mid T = Tx) \mathbb{1}_C(x) h_\theta(Tx) d\nu^*(x) \\
&= \int_{\mathcal{X}} \mathbb{1}_A(x) \mathbb{1}_C(x) h_\theta(Tx) d\nu^*(x) \\
&= \int_{\mathcal{X}} \mathbb{1}_A(x) \mathbb{1}_C(x) dP_\theta(x) \\
&= P_\theta(A \cap C)
\end{aligned}$$

which completes the proof by Proposition C.9. □

By observing the proofs so far, we see that to check sufficiency in the Bayesian sense it is enough to check only those prior distributions that are discrete (i.e., those μ_Θ on Θ where $\mu(C) = 1$ for some at most countable subset $C \subset \Theta$).

Corollary 3.12. Assuming the nice condition of Definition 1.5 . We have that $T : \mathcal{X} \to \mathcal{T}$ is sufficient (in either classical of Bayesian sense) if and only if the Bayesian sufficiency condition is satisfied for all discrete prior distributions μ_Θ on Θ.

Proof. Observe that the measures μ used in the proof of Lemma 3.10 are all discrete. Once ν^* is constructed, sufficiency follows as seen in the proof of the previous Theorem. $\square$

3.3. **Fisher-Neyman factorization.** There is a convenient characterization of sufficiency.

Theorem 3.13. Assume the nice condition of Definition 1.5. Then a statistic $\mathcal{T} : \mathcal{X} \to \mathcal{T}$ is sufficient if and only if the following holds: There is Borel $g : \mathcal{X} \to \mathbb{R}_{\geq 0}$ such that for all $\theta \in \Theta$ we have a Borel $h_\theta : \mathcal{T} \to \mathbb{R}_{\geq 0}$ such that

$$f(x|\theta) = h_\theta(Tx)g(x) \quad \text{for } P_\theta \text{ almost all } x \in X$$

Proof. First suppose this condition holds. Thus by Corollary 3.12 we only need to check Bayesian sufficiency for any discrete prior μ on Θ. Say μ is supported on $C = \{\theta_1, \theta_2, \ldots\} \subset \Theta$. For $C' \subset C$ we have by Bayes' formula that

$$P(\Theta \in C \mid X = x) = \frac{\int_{C'} f(x|\theta) d\mu(\theta)}{\int_\Theta f(x|\theta) d\mu(\theta)} = \frac{\sum_{\theta \in C'} h_\theta(Tx) g(x) \mu(\{\theta\})}{\sum_{\theta \in C} h_\theta(Tx) g(x) \mu(\{\theta\})} = \frac{\sum_{\theta \in C'} h_\theta(Tx) \mu(\{\theta\})}{\sum_{\theta \in C} h_\theta(Tx) \mu(\{\theta\})}.$$

Observe now that this is $\mathcal{T}$-measurable since the sums are countable, thus Bayesian sufficiency is established. (**Remark:** this is the reason we restrict without loss of generality to μ discrete, as otherwise it seems one cannot show measurability in the integrand unless we also assume that $(\theta, t) \mapsto h_\theta(t)$ is Borel, which we do not want to as we have not been able to verify this in Lemma 3.10).

Conversely, suppose that $\mathcal{T}$ is sufficient. Thus by Lemma 3.10 we have

$$f(x|\theta) = \frac{dP_\theta}{d\nu}(x) = \frac{dP_\theta}{d\nu^*}(x)\frac{d\nu^*}{d\nu}(x) = h_\theta(Tx)g(x)$$

for ν^*-almost all $x \in X$. As $P_\theta \ll \nu^*$ we have it also holds for P_θ-almost all $x \in X$. $\square$

Example 3.14. In Example 3.2 we have that

$$f(x|\theta) = P_\theta(X = x) = \theta^{T(x)}(1-\theta)^{n-T(x)}$$

so can just take

$$h_\theta(t) = \theta^t(1-\theta)^{n-t}$$

and $g(x) = 1$. Here $\nu(\{x\}) = 1$ is the counting measure. If instead we had a different measure $\nu(\{x\}) = q(x) > 0$ then we would have

$$f(x|\theta) = P_\theta(X = x)q(x)^{-1} = h_\theta(Tx)q(x)^{-1}$$

so then $g(x) = q(x)^{-1}$. This shows that $g(x)$ is just a change of density.

Example 3.15. Let $\mathcal{X} = \mathbb{R}^n$, $\Theta = \{(\mu, \sigma) \in \mathbb{R}^2 \mid \mu \in \mathbb{R}, \sigma > 0\}$ and for $\theta = (\mu, \sigma)$ let P_θ be the distribution of $(X_1, \ldots, X_n)$ where each X_i is Gaussian with mean μ and variance σ^2 and the X_i are independent. Thus if ν is the Lebesgue measure on $\mathbb{R}^d$ then

$$\frac{dP_\theta}{d\nu}(x_1, \ldots, x_n) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2\right).$$

Now let $\overline{x} = \frac{1}{n}\sum_{i=1}^n x_i$ for $x = (x_1, \ldots, x_n)$ and

$$s^2 = \frac{1}{(n-1)}\sum_{i=1}^n (x_i - \overline{x})^2.$$

Thus the sum inside the exponential can be written as

$$\begin{aligned}\sum_{i=1}^{n}(x_i-\overline{x}+\overline{x}-\mu) &= \sum_{i=1}^{n}(x_i-\overline{x})^2+2(\overline{x}-\mu)\sum_{i=1}^{n}(x_i-\overline{x})+\sum_{i=1}^{n}(\overline{x}-\mu)^2. \\ &= (n-1)s^2+2(\overline{x}-\mu)(n\overline{x}-n\overline{x})+\sum_{i=1}^{n}(\overline{x}-\mu)^2. \\ &= (n-1)s^2+\sum_{i=1}^{n}(\overline{x}-\mu)^2.\end{aligned}$$

Thus we have that

$$\frac{dP_\theta}{d\nu}(x_1,\dots,x_n)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}\left((n-1)s^2+\sum_{i=1}^{n}(\overline{x}-\mu)^2\right)\right),$$

which may be written as $h_\theta(T(x))$ for some function h_θ where

$$T(x)=(\overline{x},s^2).$$

Thus T, which gives the sample average and sample variance, is a sufficient statistic.

3.4. **Conditional Expectations with respect to sufficient statistics.** Suppose $T:\mathcal{X}\to\mathcal{T}$ is sufficient for the family P_θ. If $\psi:\mathcal{X}\to\mathbb{C}$ is a Borel-measurable function such that

$$E_\theta(|\psi|):=\int_{\mathcal{X}}|\psi(x)|dP_\theta(x)<\infty \quad \text{for all } \theta\in\Theta,$$

then the conditional expectation $E_\theta(\psi\mid T=t)$ is also L^1. We would like to define a canonical version for this conditional expectation via

$$E_\theta(\psi\mid T=t)=\int_{\mathcal{X}}\psi(x)d\nu_t(x) \tag{4}$$

where $\nu_t(\cdot)=P_\theta(\cdot\mid T=t)$ are the measures as defined in the definition of classical sufficiency. The issue is that the integral on the right hand side of (4) may not be absolutely convergent for some $t\in\mathcal{T}$. However, it is on some set $\mathcal{T}_\psi\subset\mathcal{T}$ that is Θ-co-null (and depends on ψ). To see this, note that

$$\int_{\mathcal{X}}|\psi(x)|dP_\theta(x)=\int_{\mathcal{X}}\int_{\mathcal{X}}|\psi(x')|d\nu_{Tx}(x')dP_\theta(x)$$

by definition of conditional probabilities (applied to positive functions) and thus if define

$$\mathcal{T}_\psi=\left\{t\in\mathcal{T}\ :\ \int_{\mathcal{X}}|\psi(x')|d\nu_t(x')<\infty\right\}$$

then this is a Borel measurable set (whose definition does not depent on θ) with $P_\theta(T^{-1}\mathcal{T}_\psi)=1$.

3.5. **Applications to estimation (Rao-Blackwell and Lehmann-Scheffe theorems).** Recall that $d:\mathcal{X}\to\mathcal{A}$ is the decision function and $L:\Theta\times\mathcal{A}\to[0,\infty)$ denotes the loss function. The risk is then

$$R(\theta,d)=\int_{\mathcal{X}}L(\theta,d(x))dP_\theta(x).$$

Recall that if $T:\mathcal{X}\to\mathcal{T}$ is a sufficient statistic then there are families of measures ν_t, for $t\in\mathcal{T}$, such that

$$P_\theta(B\mid T=t)=\nu_t(B)\quad \text{for all } B\subset\mathcal{X}.$$

Theorem 3.16 (Rao-Blackwell theorem)**.** Suppose that $\mathcal{A} = \mathbb{R}^d$ and that for each fixed θ, the map $a \mapsto L(\theta, a)$ is convex. Suppose also that $x \mapsto d(x)$ is an L^1 function for each P_θ, in other words

$$\int_{\mathcal{X}} \|d(x)\| dP_\theta < \infty.$$

Suppose that $T : \mathcal{X} \to \mathcal{T}$ is a sufficient statistic for the family of P_θ. Then there is a T-measurable $\mathcal{X}_d \subset \mathcal{X}$ such that $P_\theta(\mathcal{X}_d) = 1$ for all $\theta \in \Theta$ and a well-defined decision rule $\hat{d} : \mathcal{X}_d \to \mathbb{R}^d$ given by

$$\hat{d}(x) = E_\theta(d|T = Tx) = \int_{\mathcal{X}} d(x')d\nu_{Tx}(x') \tag{5}$$

which satisfies

$$R(\theta, \hat{d}) \leq R(\theta, d).$$

Well-defined here means that integral is absolutely convergent for all $x \in \mathcal{X}_d$.

Note that we can extend $\hat{d}$ to all of $\mathcal{X}$ and the risk functions will not change.

Proof. Note that as d is L^1 with respect to P_θ, we can apply our canonical formula (4) with $d = \psi$. Thus we can let $\mathcal{X}_d$ be the set of all x so that the integral converges absolutely (it does not depend on θ) and we see from the discussion below (4) that $P_\theta(\mathcal{X}_d) = 1$ for all $\theta \in \Theta$. Thus the expression for $\hat{d}(x)$ is well defined. By Jensen's theorem, we have that

$$L(\theta, \hat{d}(x)) = L\left(\theta, \int_{\mathcal{X}} d(x')d\nu_{Tx}(x')\right) \leq \int_{\mathcal{X}} L(\theta, d(x'))d\nu_{Tx}(x').$$

Now we integrate both sides with respect to P_θ to get that

$$\begin{aligned} R(\theta, \hat{d}) &\leq \int_{\mathcal{X}} \int_{\mathcal{X}} L(\theta, d(x'))d\nu_{Tx}(x')dP_\theta(x) \\ &= \int_{\mathcal{X}} L(\theta, d(x))dP_\theta(x) \\ &= R(\theta, \hat{d}) \end{aligned}$$

where we used the law of total probability applied to $P_\theta(\cdot \mid T = Tx) = \nu_{Tx}(\cdot)$. $\square$

Thus the Rao-Blackwell theorem allows us to construct more efficient estimators that are a function of the sufficient statistic T.

Examples of such convex loss functions include the square loss

$$L(\theta, a) = |\theta - a|^2.$$

Assume now for the rest of this section that the action space is $\mathcal{A} = \mathbb{R}$.

Definition 3.17. Let $c : \Theta \to \mathbb{R}$. An *unbiased estimator of* c is a decision rule $d : X \to \mathbb{R}$ such that

$$E_\theta(d(X)) = c(\theta) \quad \text{for all } \theta \in \Theta.$$

In the case that $E_\theta|d(X)|^2 < \infty$ for all $\theta \in \Theta$, we say that d is a UMVU (uniformly minimum variance unbiased) estimator of c if for any other unbiased estimator $d_2 : X \to \mathbb{R}$ of c with $E_\theta|d_2(X)|^2 < \infty$ for all $\theta \in \Theta$ we have that

$$E_\theta|d(X) - c(\theta)|^2 \leq E_\theta|d_2(X) - c(\theta)|^2 \quad \text{for all } \theta \in \Theta.$$

Note that if d is an unbiased estimator of c that is L^1 w.r.t each P_θ and T is sufficient, then the estimator $\hat{d}$ in the Rao-Blackwell theorem (given by (5)) is also unbiased since

$$E_\theta(\hat{d}(X)) = \int_{\mathcal{X}} E_\theta(d|T = Tx)dP_\theta(x) = \int_{\mathcal{X}} d(x)dP_\theta(x) = c(\theta).$$

The Rao-Blackwell theorem applied to the square loss shows that if d is UMVU then so is $\hat{d}$.

We let $\mathcal{B}_T$ denote the σ-algebra on $\mathcal{X}$ induced by the statistic $T : \mathcal{X} \to \mathcal{T}$.

Proposition 3.18. Suppose that $d : \mathcal{X} \to \mathbb{R}$ is a UMVU esimator of $c : \Theta \to \mathbb{R}$. Then if T is a sufficient statistic, then $d \in L^2(\mathcal{X}, \mathcal{B}_T, P_\theta)$ for all $\theta \in \Theta$. In other words, d is $\mathcal{B}_T$ almost measurable and so $d(x) = \hat{d}(x)$ for P_θ-almost all $x \in \mathcal{X}$.

Proof. Fix $\theta \in \Theta$. Note that $\hat{d} = E_\theta(d|T)$ is the orthogonal projection of d onto the subspace $L^2(\mathcal{X}, \mathcal{B}_T, P_\theta)$, thus $d - \hat{d}$ is orthogonal to every element on this subspace. Note that $c(\theta)$, being a constant function on $\mathcal{X}$, is also in this subspace. Thus $d - \hat{d}$ and $\hat{d} - c(\theta)$ are orthogonal and so the Pythagoras theorem gives

$$E_\theta(d - c(\theta))^2 = E_\theta(d - \hat{d})^2 + E_\theta(\hat{d} - c(\theta))^2. \tag{6}$$

However, as d is UMVU, so must be $\hat{d}$ and they must achieve the same error from $c(\theta)$. Thus by (6) we have that

$$E_\theta(d - \hat{d})^2 = 0$$

and so $d = \hat{d}$ holds P_θ-almost surely. □

We now want some conditions that imply that $\hat{d}$ is an actual UMVU. It turns out the concept of a complete sufficient statistic gives us this.

Definition 3.19 (Complete sufficient statistic)**.** We say that $T : \mathcal{X} \to \mathcal{T}$ is a *complete sufficient statistic* if T is sufficient and satisfies the following property: Whenever a measurable $g : \mathcal{T} \to \mathbb{R}$ satisfies that

$$E_\theta|g \circ T| < \infty \text{ and } E_\theta(g \circ T) = 0 \quad \text{for all } \theta \in \Theta$$

then we must have that

$$g \circ T = 0 \quad P_\theta\text{-almost surely for all } \theta \in \Theta.$$

Here is a geometric way of viewing completeness (at least for square measurable g): It says that if $g \circ T \in L^2(\mathcal{X}, \mathcal{B}_T, P_\theta)$ is orthogonal to the space of constant functions, for all $\theta \in \Theta$, then $g \circ T$ must be the zero vector in $L^2(\mathcal{X}, \mathcal{B}_T, P_\theta)$, for all $\theta \in \Theta$. We stress that it is not enough for the implication to hold for every fixed $\theta \in \Theta$ (i.e., we need the orthogonality to hold for all $\theta \in \Theta$ before we conclude that $g \circ T = 0$.).

Theorem 3.20 (Lehmann-Scheffe theorem)**.** Suppose that T is a complete sufficient statistic. Suppose that $d : \mathcal{X} \to \mathbb{R}$ is an unbiased estimator of $c : \Theta \to \mathbb{R}$ with $E_\theta|d|^2 < \infty$ for all $\theta \in \Theta$. Then $\hat{d} = E_\theta(d|T)$ is the unique UMVU of c.

Proof. Let us first show that $\hat{d}$ is a UMVU before we show uniqueness. Suppose for contradiction that $d_2 : \mathcal{X} \to \mathbb{R}$ is another unbiased estimator, with $E_\theta|d_2|^2 < \infty$ for all $\theta \in \Theta$ such that, for some $\theta_0 \in \Theta$ we have that

$$E_{\theta_0}|d_2 - c(\theta)|^2 < E_{\theta_0}|\hat{d} - c(\theta)|^2.$$

Now by applying the Rao-Blackwell theorem to d_2, we have that $\hat{d}_2$ is an even better unbiased estimator for $c(\theta)$, and thus

$$E_{\theta_0}|\hat{d}_2 - c(\theta)|^2 < E_{\theta_0}|\hat{d} - c(\theta)|^2.$$

This implies that if we define $g : \mathcal{T} \to \mathbb{R}$ by[2] $g(Tx) = \hat{d}(x) - \hat{d}_2(x)$, then we must have that $g \circ T$ is not the zero-function in $L^2(\mathcal{X}, \mathcal{B}_T, P_{\theta_0})$. However, we have for all $\theta \in \Theta$ that

$$E_\theta(g \circ T) = E_\theta(\hat{d} - \hat{d}_2) = c(\theta) - c(\theta) = 0,$$

which contradicts the completeness of T. Thus $\hat{d}$ is indeed a UMVU.

Let us now show that it is unique. By Proposition 3.18 we have that if $d_3 : \mathcal{X} \to \mathbb{R}$ is another UMVU, then $d_3 = \hat{d}_3 \in L_2(X, \mathcal{B}_T, P_\theta)$ but $\hat{d}(x) - \hat{d}_3(x) =: g(Tx)$ is a function of T. Thus

$$E_\theta(\hat{d} - d_3) = E_\theta(g \circ T) = c(\theta) - c(\theta) = 0$$

for all $\theta \in \Theta$ and so by definition of completeness we must have that $\hat{d}(x) = \hat{d}_3(x) = d_3(x)$ holds P_θ-almost surely. □

Example 3.21. Let $\mathcal{X} = \{0,1\}^n$, $\Theta = [0,1]$ and suppose that P_θ is the distribution of $(X_1, \ldots, X_n)$ where X_i is Bernoulli with $P(X_i = 1) = \theta$ and $X_1, \ldots, X_n$ are i.i.d. Let us now show that $T(x_1, \ldots, x_n) = \sum_{i=1}^n x_i$ is a complete sufficient statistic. Suppose that $g : \mathbb{Z}_{\geq 0} \to \mathbb{R}$ is any function (we do not need any L^1 assumptions as $\mathcal{X}$ is finite), then

$$E_\theta g(Tx) = \sum_{t=0}^n g(t)\binom{n}{t}\theta^t(1-\theta)^{n-t}.$$

The right hand side is a polynomial in θ, thus if it is zero for all $\theta \in [0,1]$ then it must be equal to the zero polynomial. We must show that if this happens, then $g(t) = 0$ for all t. Suppose this it not the case, thus there is a minimal $k \in \{0, \ldots, n\}$ so that $g(k) \neq 0$. Hence

$$E_\theta g(Tx) = \sum_{t=k}^n g(t)\binom{n}{t}\theta^t(1-\theta)^{n-t}.$$

The coefficient of θ^k can only have a contribution form the $t = k$ term, as for $k < t \leq n$ we have that $\theta^t(1-\theta)^{n-t}$ is a polynomial with a factor θ^t. So the coefficient of θ^k is in fact $g(k)\binom{n}{k}\theta^k$ and so if this coefficient is zero then we must have $g(k) = 0$, a contradiction. Hence we have shown that if $E_\theta(g \circ T) = 0$ for all $\theta \in [0,1]$ (in fact, at least $n+1$ such θ will do), then $g \circ T = 0$, thus showing completeness (sufficiency was already shown in a previous example).

Now applying the Lehmann-Scheffe theorem, we have that $d(x) = \frac{1}{n}\sum_{i=1}^n x_i$ is a UMVU for θ as it can be written as $d(x) = \frac{1}{n}T(x)$ and so $\hat{d} = d$. This loss is

$$E_\theta(\hat{d} - \theta)^2 = Var(\frac{1}{n}(X_1 + \cdots + X_n)) = \frac{1}{n}Var(X_1) = \frac{1}{n}\theta(1-\theta).$$

Definition 3.22. We say that $c : \Theta \to \mathbb{R}$ is *estimatable* if there exists a $d : \mathcal{X} \to \mathbb{R}$ that is an unbiased estimator of c, i.e., d is integrable w.r.t each P_θ and $E_\theta d = c(\theta)$.

Note that the space of estimatable functions forms a vector space.

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[2]This is well defined as $\hat{d}, \hat{d}_2$ are functions of T by construction in the Rao-Blackwell theorem above.

Example 3.23. Let us continue with the Example 3.21 and try to find all possible estimatable functions $c : \Theta \to \mathbb{R}$. Suppose that c is estimatable by d, we can assume $d = d(T)$ is a function of T (by replacing d with $\hat{d}$). Thus

$$c(\theta) = \sum_{t=0}^{n} d(t) \binom{n}{t} \theta^t (1-\theta)^{n-t}$$

is in fact a polynomial in θ of degree at most n. Thus the space of estimatable functions is at most $n+1$ dimensional. Now consider

$$d((x_1, \ldots, x_n)) = X_1 \ldots X_k$$

for $k \in \{0, \ldots, n\}$. We have

$$E(d) = E(X_1 \cdots X_k) = E(X_1)^k = \theta^k.$$

Thus the estimatable functions are exactly the polynomials in θ of degree at most n.

Example 3.24. We now construct distributions P_θ which have no complete sufficient statistic. Let $\mathcal{X} = \Theta = \mathbb{Z}$ and let P_θ be the uniform distribution on $\{\theta, \theta+1\}$, i.e., $P_\theta(\theta) = \frac{1}{2} = P_\theta(\theta+1)$. Now suppose that $T : \mathcal{X} \to \mathcal{T}$ is a sufficient statistic. We claim that T must be injective. First, by Fischer-Neyman Factorization theorem, we have that

$$P_\theta(x) = h_\theta(Tx)g(x)$$

for some functions h_θ and g. Suppose for contradiction that $Ta = Tb = t$ for some integers $a < b$. Now

$$\frac{1}{2} = P_x(x) = h_x(Tx)g(x)$$

and so $g(x) \neq 0$ for all $x \in \mathbb{Z}$. Observe that

$$0 = P_b(a) = h_b(Ta)g(a) = h_b(t)g(a)$$

and thus $h_b(t) = 0$ as $g(a) \neq 0$. On the other hand

$$\frac{1}{2} = P_b(b) = h_b(Tb)g(b) = h_b(t)g(b) = 0,$$

which is a contradiction. Thus T must be injective. This means that any function $G : \mathcal{X} \to \mathbb{R}$ is a function of T. In particular, the function $G(x) = (-1)^x$ is a function of T. But

$$E_\theta G = 0 \quad \text{for all } \theta \in \Theta$$

but G is not the zero function, thus this shows that T can not be complete. Note that the identity function $T(x) = x$ is sufficient, as is always the case.

3.6. **Exponential Families.** We now construct a rich class of parametric families P_θ which have complete sufficient statistics.

Definition 3.25. A k-parameter exponential family is a family of distributions P_θ, for $\theta \in \Theta$ on $\mathcal{X}$ such that, with respect to some σ-finite measure ν on $\mathcal{X}$ we have that $P_\theta \ll \nu$ for all $\theta \in \Theta$ with density functions given by

$$p(x|\theta) := \frac{dP_\theta}{d\nu}(x) = C(\theta)h(x)\exp\left(\sum_{i=1}^{k} \pi_i(\theta)T_i(x)\right)$$

for some measurable $\pi_i : \Theta \to \mathbb{R}$, $T_i : \mathcal{X} \to \mathbb{R}$, $h : \mathcal{X} \to \mathbb{R}_{\geq 0}$ such that

$$C(\theta) = \left(\int_{\mathcal{X}} h(x)\exp\left(\sum_{i=1}^{k} \pi_i(\theta)T_i(x)\right) d\nu(x)\right)^{-1} > 0$$

is a positive finite normalization constant for all $\theta \in \Theta$, i.e., we assume that this integral is absolutely convergent and has a non-zero value.

Note that $T : \mathcal{X} \to \mathbb{R}^k$ given by $T(x) = (T_1(x), \dots, T_k(x))$ is a sufficient statistic, by the Fischer-Neyman factorization.

We let $\pi : \Theta \to \mathbb{R}^k$ be the map given by $\pi(\theta) = (\pi_1(\theta), \dots, \pi_k(\theta))$.

Proposition 3.26. Suppose that the range of π, i.e., the set

$$\pi(\Theta) = \{\pi(\theta) \in \mathbb{R}^k \mid \theta \in \Theta\}$$

contains a non-empty open set. Then $T : \mathcal{X} \to \mathbb{R}^k$ is a complete sufficient statistic for the P_θ.

Proof. We must show that if $g : \mathbb{R}^k \to \mathbb{R}$ is measurable such that

$$E_\theta |g \circ T| < \infty \text{ and } E_\theta(g \circ T) = 0$$

for all $\theta \in \Theta$ then $g(Tx) = 0$ for P_θ-almost all $x \in \mathcal{X}$. Note that

$$E_\theta(g \circ T) = \int_{\mathbb{R}^k} g(u) d(T^* P_\theta)(u)$$

and thus we will now compute the pushforward measure $T^* P_\theta$. Let ν_T be the pushforward measure of $h(x)d\nu(x)$, thus

$$E_\theta(g \circ T) = \int_{\mathcal{X}} C(\theta) h(x) g(Tx) \exp\left(\sum_{i=1}^{k} \pi_i(\theta) T_i\right) d\nu(x) = \int_{\mathbb{R}^d} C(\theta) g(u) \exp\left(\langle \pi(\theta), u\rangle\right) d\nu_T(u)$$

where all these integrals are absolutely convergent by assumption.

Now let μ_θ be the finite signed measure on $\mathbb{R}^k$ with density given by

$$\frac{d\mu_\theta}{d\nu_T}(u) = C(\theta) g(u) \exp\left(\langle \pi(\theta), u\rangle\right).$$

Note that μ_θ is a finite signed measure since the density is L^1 w.r.t ν_T by assumption.

Thus we have

$$0 = E_\theta(g \circ T) = \int_{\mathbb{R}^k} 1 \mu_\theta(u)$$

As $\pi(\Theta)$ is open, choose δ so small so that $\pi(\theta) + v \in \pi(\Theta)$ for all $\|v\| < \delta$. Fixing such v, we can thus write $\pi(\theta) + v = \pi(\theta_v)$ for some $\theta_v \in \theta$.

Now observe that

$$\begin{aligned} E_{\theta_v}(g \circ T) &= C(\theta_v) \int_{\mathbb{R}^d} g(u) \exp(\langle \pi(\theta_v), u\rangle) d\nu_T(u) \\ &= C(\theta_v) \int_{\mathbb{R}^d} g(u) \exp(\langle \pi(\theta) + v, u\rangle) d\nu_T(u) \\ &= C(\theta)^{-1} C(\theta_v) \int_{\mathbb{R}^d} \exp(\langle v, u\rangle) d\mu_\theta(u) \end{aligned}$$

with all integrals absolutely convergent by assumption. As $E_{\theta_v}(g \circ T) = 0$ we have

$$\int_{\mathbb{R}^k} \exp(\langle v, u\rangle) d\mu_\theta = 0$$

with this integral being absolutely convergent. But this integral is exactly equal to the moment generating function $M_{\mu_\theta}(v)$. Thus we have shown that the moment generating function of μ_θ is defined and equals 0 on a neighbourhood of the origin of $0 \in \mathbb{R}^k$, thus $\mu_\theta = 0$ by Theorem G.11.

Thus $g(u) = 0$ for ν_T-almost all $u \in \mathbb{R}^k$. That is

$$0 = \int |g(u)|\nu_T = \int |g(Tx)|h(x)d\nu(x)$$

and thus $g(Tx) = 0$ holds for P_θ-almost all $x \in \mathcal{X}$. $\square$

Example 3.27. We continue from Example 3.15, where we recall that $\mathcal{X} = \mathbb{R}^n$, $\Theta = \{(\mu, \sigma) \in \mathbb{R}^2 \mid \mu \in \mathbb{R}, \sigma > 0\}$ and for $\theta = (\mu, \sigma)$ let P_θ be the distribution of $(X_1, \ldots, X_n)$ where each X_i is Gaussian with mean μ and variance σ^2 and the X_i are independent. Thus if ν is the Lebesgue measure on $\mathbb{R}^d$ then

$$\frac{dP_\theta}{d\nu}(x_1, \ldots, x_n) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2\right).$$

We saw that we can rewrite this as

$$\frac{dP_\theta}{d\nu}(x_1, \ldots, x_n) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-n\mu}{2\sigma^2}\right) \exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^n x_i^2 + \frac{1}{\sigma^2}\mu\sum_{i=1}^n x_i\right).$$

Thus we see that the P_θ form a 2-parameter exponential family with

$$T_1(x) = \sum_{i=1}^n x_i^2, \quad T_2(x) = \sum_{i=1}^n x_i$$

and

$$\pi(\mu, \sigma^2) = (-\sigma^{-2}/2, \mu\sigma^{-2}).$$

Thus the image of π is the set $(-\infty, 0) \times (0, \infty)$ which is an open subset of $\mathbb{R}^2$, thus by the theorem above $T(x) = (T_1(x), T_2(x))$ is a sufficient statistic.

Note that the sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^n (x_i - \overline{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^n x_i^2 + (n-1)\overline{x}^2\right)$$

is a function of $T(x)$ and thus the more common statistic $x \mapsto (\overline{x}, s^2)$, which was discussed in Example 3.15, is also a complete sufficient statistic (as any function of this statistic is also a function of T, completeness is inherited from T).

Now what if we defined $\Theta' = \{(\mu, \sigma) \mid \mu = 0, \sigma > 0\}$. Then we will still have an exponential family and T will still be sufficient, but the image of π is now no longer an open subset of $\mathbb{R}^2$, so we cannot conclude that T is a complete statistic. Indeed, it is not since if we define

$$g(x) = \sin\left(\sum_{i=1}^n x_i\right)$$

then we see that g is indeed a function of T but $g(x)\frac{dP_\theta}{d\nu}(x)$ is an odd function of x and thus $\int g(x)dP_\theta(x) = 0$. However clearly g is not the zero function. Thus T is not complete for the subfamily Θ'.

Example 3.28. Let $\mathcal{X} = \{0, 1\}^n$, $\Theta = (0, 1)$ and let P_θ be the distribution on $\mathcal{X}$ where $(X_1, \ldots, X_n) \in \mathcal{X}$ has i.i.d component with $P_\theta(X_i = 1) = \theta$. Thus, letting $T(x) = \sum_{i=1}^n x_i$ we have

$$P_\theta(x_1, \ldots, x_n) = \theta^{T(x)}(1-\theta)^{n-Tx} = (1-\theta)^n(\theta(1-\theta)^{-1})^{T(x)} = C(\theta)\exp(\log(\theta(1-\theta)^{-1})T(x)).$$

Thus this forms a 1-paremeter exponential family with

$$\pi(\theta) = \log\left(\frac{\theta}{1-\theta}\right).$$

The image of π is thus $\mathbb{R}$, which is an open subset of $\mathbb{R}$. Thus T is a complete and sufficient statistic, as we saw earlier in a previous example.

4. Hypothesis Testing

We recall the concepts of Hypothesis Testing introduced in Definition 1.3. Suppose we have a partition $\Theta = \Theta_0 \sqcup \Theta_1$ of the parameter space Θ. Our action space is $\mathcal{A} = \{\Theta_0, \Theta_1\}$ and we have a decision rule, which we call a *non-randomized test*, $d : \mathcal{X} \to \{\Theta_0, \Theta_1\}$ and our loss function is

$$L(\theta, \Theta_i) = 1 - \mathbb{1}_{\Theta_i}(\theta).$$

Thus $L(\theta, d(x))$ equals 1 if $\theta \in d(x)$ and 0 if $\theta \notin d(x)$.

Interpretation: The parameter θ is unknown to the statistician, who samples $x \in \mathcal{X}$ according to P_θ. The statistictian must now *guess wisely* based on the sampeld x whether θ is in Θ_0 or Θ_1, thus $d(x) \in \{\Theta_0, \Theta_1\}$ represents the statisticians choice. The case $\theta \in \Theta_0$ is called the *null hypothesis* while the case $\theta \in \Theta_1$ is called the *alternative hypothesis.* So we say that the statistician *rejects the null hypothesis* if $d(x) = \Theta_1$, while the statiscitian *accepts the null hypothesis* if $d(x) = \Theta_0$.

The associated risk is

$$R(\theta, d) = \int_{\mathcal{X}} L(\theta, d(x)) = P_\theta(\theta \notin d(x)),$$

i.e., the probability of choosing the wrong hypothesis.

Definition 4.1. A *randomized test* is a map $\varphi : \mathcal{X} \to [0,1]$. Define maps

$$\alpha_\varphi : \Theta_0 \to [0,1]$$

by

$$\alpha_\varphi(\theta) = E_\theta(\varphi)$$

and $\beta_\varphi : \Theta_1 \to [0,1]$ given by

$$\beta_\varphi(\theta) = 1 - E_\theta(\varphi).$$

Interpretation: A *randomized* test is a decision rule where, after x is sampled, the value $d(x)$ is randomly chosen where with probability $\varphi(x)$ we choose $d(x) = \Theta_1$ and with probability $1 - \varphi(x)$ we choose $d(x) = \Theta_0$. Thus for $\theta \in \Theta_0$, we have that $\alpha_\varphi(\theta)$ is the probability that we falsely reject the null hypothesis, that is, we falsely choose the hypothesis Θ_1 where firstly x is chosen randomly according to P_θ and then $d(x) = \Theta_i$ is chosen randomly where $P(d(x) = \Theta_1 \mid x) = \varphi(x)$. Hence

$$P_\theta(d(x) = \Theta_1) = \int_{\mathcal{X}} P(d(x) = \Theta_1 \mid x) dP_\theta(x) = E_\theta(\varphi),$$

which explains how $\varphi(x)$ is interpreted as a probability. A non-randomized rule is the case where $\varphi(x) \in \{0,1\}$ for all $x \in \mathcal{X}$.

Definition 4.2. The *size* of a randomized test φ is equal to

$$\alpha_\varphi = \sup_{\theta \in \Theta_0} \alpha_\varphi(\theta) \in [0,1].$$

The *power* of φ is $\gamma_\varphi : \Theta_1 \to [0,1]$ is defined as

$$\gamma_\varphi(\theta) = E_\theta(\varphi).$$

Interpretation: The size of the test is the supremum probability of falsely rejecting the null hypothesis. So it is the least upper bound (over $\theta \in \Theta_0$) that measures the probability that we falsely reject the null hypothesis (that is, we choose Θ_1) when in fact the null hypothesis ($\theta \in \Theta_0$) is true. The power measures the probabiliy that we can get a strong enough signal to reject the null hypothesis, in the case that the null hypothesis is indeed actually false.

So we want to minimize the size, but maximise the power. Our number one priority is to keep the size low, thus the null hypothesis should be rejected only if there is strong evidence against it. Thus we should let the null hypothesis be something we wish to prove false. Typically, we want $\alpha \leq 0.05$.

Example 4.3. Suppose that a manufacturor wishes to test whether a product has a defect or not. Suppose θ is the (unknown) proportion of products that have a defect. Say the manufacturer wishes that $\theta \leq 0.05$. Thus we let the null hypothesis be $\theta \in \Theta_0 = (0.05, 1]$ (as we wish to prove it wrong) and the alternative hypothesis is $\theta \in \Theta_1 = [0, 0.05)$. The manufacturer randomly samples 40 such products so the sample is $x = (x_1, \ldots, x_{40}) \in \mathcal{X} = \{0,1\}^{40}$ (where $x_i = 1$ means the ith product in the sample has a defect) and has a decision rule given by $d_1(x) = \Theta_1$ if and only if there was at most 1 defect. In other words, we have a non-randomized test given by

$$\varphi_1(x) = \mathbb{1}(\sum_{i=1}^{40} x_i \leq 1).$$

Thus

$$\alpha_{\varphi_1}(\theta) = P_\theta(\sum_i x_i \leq 1) = (1-\theta)^{40} + 40(1-\theta)^{39}\theta$$

for $\theta \in \Theta_0 = (0.05, 1]$. Some simple calculus shows that the supremum is attained at the endpoint $\theta = 0.05$. Thus the size of the test is

$$\alpha_{\varphi_1} = 0.399.$$

This means that the statistician cannot confidently reject the hypotesis $\theta > 0.05$, even if at most 5% of the sample has a defect. He simply does not have enough data to make $\alpha_{\varphi_1} \leq 0.05$.

Now if instead he takes 100 samples, so $\mathcal{X} = \{0,1\}^{100}$ and uses the same criteria of only at most 1 defect, then we get that

$$\alpha_\varphi = 0.95^{100} + 100(0.95)^{99} \times 0.05 = 0.037$$

and so the test has a good enough size in this case. For $\theta = 0.05$, the power $\gamma_\varphi(\theta)$ of this test is equal to α, but as $\theta \to 0$ then the power converges to 1, i.e., if θ is close to 0 then with very high probability we (correctly) reject the null hypothesis.

4.1. **Neyman-Pearson Lemma.** Say we want to choose a test φ where α_φ is at most some fixed value, usually 0.05. There are many such tests to choose from, but we would like those which have the highest possible power. This motivates the following definition.

Definition 4.4 (UMP test)**.** A *UMP (uniformly most powerful) test of size* $\alpha \in [0,1]$ is a test φ such that $\alpha_\varphi = \alpha$ and for all tests φ' with $\alpha_{\varphi'} \leq \alpha$ we have

$$\gamma_\varphi(\theta) \geq \gamma_{\varphi'}(\theta) \quad \text{for all } \theta \in \Theta_1.$$

We recall that we always assume the nice condition of Definition??, i.e.,

$$\frac{dP_\theta}{d\nu} = f(x|\theta)$$

where ν is a σ-finite measure on $\mathcal{X}$ and $(x,\theta) \mapsto f(x|\theta)$ is measurable.

Proposition 4.5 (Neyman-Pearson for simple hypotheses)**.** Suppose that $\Theta = \{\theta_0, \theta_1\}$ has only two points and let the null hypothesis be $\Theta_0 = \{\theta_0\}$ and the alternative is $\Theta_1 = \{\theta_1\}$. Then for each constant $k \in [0, \infty)$ and $\gamma \in [0, 1]$ consider the test

$$\varphi(x) = \begin{cases} 1, & \text{if } f(x|\theta_1) > kf(x|\theta_0) \\ \gamma, & f(x|\theta_1) = kf(x|\theta_0) \\ 0, & f(x|\theta_1) < kf(x|\theta). \end{cases}$$

Then φ is a UMP test of size

$$\alpha = P_{\theta_0}\left(f(x|\theta_1) > kf(x|\theta_0)\right) + \gamma P_{\theta_0}\left(f(x|\theta_1) = kf(x|\theta_0)\right).$$

Conversely, given any $\alpha \in (0, 1)$, we can find such k, γ so that the test φ is UMP of size α.

Proof. Suppose that ψ is a test of size at most $\alpha_\psi \leq \alpha$. The power difference is

$$\begin{aligned} \gamma_\varphi(\theta_1) - \gamma_\psi(\theta_1) &= E_{\theta_1}\varphi - E_{\theta_1}\psi \\ &= \int_{A_1} (\varphi(x) - \psi(x))f(x|\theta_1)d\nu(x) + \int_{A_2} (\varphi(x) - \psi(x))f(x|\theta_1)d\nu(x) + \int_{A_3} (\varphi(x) - \psi(x))f(x|\theta_1)d\nu(x) \end{aligned}$$

where

$$\begin{aligned} A_1(x) &= \{x \in \mathcal{X} \mid f(x|\theta_1) > kf(x|\theta_0)\}, \\ A_2(x) &= \{x \in \mathcal{X} \mid f(x|\theta_1) = kf(x|\theta_0)\}, \\ A_3(x) &= \{x \in \mathcal{X} \mid f(x|\theta_1) < kf(x|\theta_0)\}. \end{aligned}$$

Now for $x \in A_1$ we have that $\varphi(x) - \psi(x) = 1 - \psi(x) \geq 0$ and $f(x|\theta_1) \geq kf(x|\theta_0)$. Thus

$$(\varphi(x) - \psi(x))f(x|\theta_1) \geq (\varphi(x) - \psi(x))f(x|\theta_1)$$

for $x \in A_1$ and hence

$$\int_{A_1} (\varphi(x) - \psi(x))f(x|\theta_1)d\nu(x) \geq \int_{A_1} (\varphi(x) - \psi(x))kf(x|\theta_0)d\nu(x).$$

For $x \in A_2$, we have $f(x|\theta_0) = kf(x|\theta_1)$ and so

$$\int_{A_2} (\varphi(x) - \psi(x))f(x|\theta_1)d\nu(x) = \int_{A_2} (\varphi(x) - \psi(x))kf(x|\theta_0)d\nu(x).$$

Now for $x \in A_3$ we have $\varphi(x) - \psi(x) = 0 - \psi(x) \leq 0$ but $f(x|\theta_1) < kf(x|\theta_0)$ and thus

$$(\varphi(x) - \psi(x))f(x|\theta_1) \geq (\varphi(x) - \psi(x))kf(x|\theta_0).$$

Hence

$$\int_{A_3} (\varphi(x) - \psi(x))f(x|\theta_1)d\nu(x) \geq \int_{A_3} (\varphi(x) - \psi(x))kf(x|\theta_0)d\nu(x).$$

Combining this inequalities we get that

$$\begin{aligned} \gamma_\varphi(\theta_1) - \gamma_\psi(\theta_1) &\geq \int_{\mathcal{X}} (\varphi(x) - \psi(x))kf(x|\theta_0)d\nu(x) \\ &= k(E_{\theta_0}\varphi - E_{\theta_0}\psi) \\ &= k(\alpha_\varphi - \alpha_\psi) \\ &\geq 0. \end{aligned}$$

Thus φ is indeed UMP.

We now show that if $\alpha \in (0,1)$, then we can choose such a k and γ. First, define

$$Y(x) = \frac{f(x|\theta_1)}{f(x|\theta_0)}.$$

Thus we can write

$$\alpha = P_{\theta_0}(Y > k) + \gamma P_{\theta_0}(Y = k).$$

Observe that P_{θ_0} almost surely, we have that $f(x|\theta_0) > 0$, so Y is well defined and $0 \le Y < \infty$. If there exists k such that $P_{\theta_0}(Y > k) = \alpha$, then we take this k and $\gamma = 0$. If not, then let

$$k_0 = \inf \{k \in \mathbb{R} \mid P_{\theta_0}(Y > k) \le \alpha\}.$$

Note that $0 \le k_0 < \infty$ since $0 \le Y < k$ almost surely.

We see that

$$P_{\theta_0}(Y > k_0) = \lim_{n\to\infty} P_{\theta_0}(Y > k_0 + 1/n) \le \alpha$$

and thus

$$P_{\theta_0}(Y > k_0) < \alpha \tag{7}$$

(as we assumed no such k exists). On the other hand

$$P_{\theta_0}(Y \ge k_0) = \lim_{n\to\infty} P_{\theta}(Y > k_0 - 1/n) \ge \alpha. \tag{8}$$

Thus $P_{\theta_0}(Y = k_0) > 0$ and so we can define

$$\gamma = \frac{\alpha - P_{\theta_0}(Y > k)}{P_\theta(Y = k)}.$$

It remains to show that $\gamma \in [0,1]$. We know $\gamma > 0$ from (7). However

$$\gamma = \frac{\alpha - P_{\theta_0}(Y > k)}{P_\theta(Y \ge k) - P_\theta(Y > k)}$$

and so from (8) we have $\gamma \le 1$. □

Example 4.6. We recall Example 1.4. Suppose we have a dice that is either fair (null hypothesis) or always rolls a 6 (alternative hypothesis). Suppose we have rolled this dice n times and observed the outcomes $(X_1, \dots, X_n) \in \{1,2,3,4,5,6\}^n = \mathcal{X}$. Say $\Theta = \{\theta_0, \theta_1\}$ where P_{θ_0} is the uniform distribution on $\mathcal{X}$ (dice is fair) while P_{θ_1} is the probability measure concentrated on $(6,\dots,6) \in \mathcal{X}$.

Let us apply the Neyman-Pearson lemma to find a UMP test. Here ν is the counting measure $\nu\{x\} = 1$ for all $x \in \mathcal{X}$. Here

$$f(x|\theta_0) = 6^{-n}$$

and

$$f(x|\theta_1) = \mathbb{1}_{\{(6,\dots,6)\}}(x).$$

Consider $0 < k < 6^n$. Now $f(x|\theta_1) > kf(x|\theta_0)$ occurs if and only if $x = (6,\dots,6)$ and otherwise we have $f(x|\theta_1) = 0 < k6^{-n} = f(x|\theta_0)$. Thus the test which rejects Θ_0 if and only if we roll all 6 is a UMP test. It has size $\alpha = 6^{-n}$ and power $P_{\theta_1}(x = (6,\dots,6)) = 1$, thus we can directly see that it is UMP.

Now consider the interesting critical case $k = 6^n$. Here we $kf(x|\theta_0) = 1$ for all x hence we will never have $f(x|\theta_1) > kf(x|\theta_0)$. So the corresponding test will accept Θ_0 if we roll something that is not a 6 but if we roll all 6 then the test will randomly, with probability $\gamma \in [0,1]$, choose to reject the null hypothesis. In other words, the test is

$$\varphi(x) = \gamma \mathbb{1}_{\{(6,\ldots,6)\}}(x).$$

Thus the size of the test is

$$\alpha = E_{\theta_0}(\varphi) = \gamma 6^{-n},$$

i.e., it is the probability of rolling all 6 and then randomly (independent of the roll) choosing to reject the null hypothesis. The power of the the test is equal to $E_{\theta_1}(\varphi) = \gamma$. By Neyman-Pearson, this test is UMP.

We have created all $\alpha \in [0, 6^{-n}]$. To get the remaining ones, we consider the case $k = 0$. We never have $f(x|\theta_1) < kf(x|\theta_0)$, while the case $f(x|\theta_1) > kf(x|\theta_0)$ occurs only if we roll all 6. We have $f(x|\theta_1) = kf(x|\theta_0)$ whenever we roll at least one non 6. In this case, we reject the null hypothesis with probability γ (which is stupid). Thus the size of this test is

$$\alpha = 6^{-n} + (1 - 6^{-n})\gamma.$$

The power of this test is 1, so clearly it is UMP.

4.2. **Unbiased tests and one-parameter exponential families.**

Definition 4.7. We say that a test $\varphi : \mathcal{X} \to [0,1]$, for a hypothesis testing problem $\Theta = \Theta_0 \sqcup \Theta_1$, is *unbiased* if the power is bounded below by the size of the test, i.e.,

$$\inf_{\theta \in \Theta_1} \gamma_\varphi(\theta) \geq \alpha_\varphi.$$

In other words

$$\inf_{\theta \in \Theta_1} E_\theta \varphi \geq \sup_{\theta \in \Theta_0} E_\theta(\varphi).$$

We say that an unbiased test φ^* is UMPU (uniformly most powerful unbiased) if for all unbiased tests φ of size $\alpha_\varphi = \alpha_{\varphi^*}$ we have that

$$\gamma_\varphi(\theta) \leq \gamma_{\varphi^*}(\theta) \quad \text{for all } \theta \in \Theta_1.$$

We now construct UMPU tests for one-parameter exponential families where $\Theta_0 = \{\theta_0\}$ contains a single point and Θ is an open subset of $\mathbb{R}$. Thus, for this section we fix density functions

$$\frac{dP_\theta}{d\nu}(x) = C(\theta)h(x)\exp(\theta T(x)).$$

Recall that these means that these densities are absolutely convergent for all θ and the integrals equal to 1.

We first show a technical lemma that states that $\theta \mapsto E_\theta(\varphi)$ is differentiable and we can differentiate under the integral sign.

Lemma 4.8. The map $\theta \mapsto C(\theta)$ is smooth (infinitely differentiable). Moreover, we can differentiate under the integral sign in the sense that

$$\frac{d}{d\theta}(E_\varphi(\theta)) = \int_{\mathcal{X}} \varphi(x) \left(C'(\theta)h(x)\exp(\theta T(x)) + C(\theta)h(x)T(x)\exp(\theta T(x)) \right) d\nu(x)$$

with this integral absolutely convergent.

Proof. To show the first claim about $C(\theta)$, we will use the standard differentiation under the integral sign theorem to show that $C(\theta)^{-1} > 0$ (defined by an integral) can be differentiated under the integral sign. This means it is enough to show that and point in Θ has open open neighbhourhood U such that

$$h(x)|T(x)|\exp(\theta T(x)) \leq K(x) \quad \text{for all } \theta \in U$$

for some integrable function $K(x) = K_U(x)$. Thus fix and $\theta_0 \in \Theta$. As Θ is open we have $\delta > 0$ so that $[\theta_0 - 2\delta, \theta_0 + 2\delta] \subset \Theta$. There exists a constant $A = A_\delta > 0$ such that

$$|t| \leq A_\delta(\exp(\delta t) + \exp(-\delta t)) \quad \text{for all } t \in \mathbb{R}.$$

Now for all $\theta \in U = (\theta_0 - \delta, \theta_0 + \delta)$ we have

$$\begin{aligned} h(x)|T(x)|\exp(\theta T(x)) &\leq A_\delta h(x)(\exp(\delta T(x)) + \exp(-\delta T(x)))\exp(\theta T(x)) \\ &= \sum_{\pm} A_\delta h(x)\exp((\theta \pm \delta)T(x)) \end{aligned}$$

But $\theta \pm \delta \in \Theta_0$ so these upper bounds are integrable. We still need to find an integrable function $K(x) = K_U(x)$ upper bounding this (independently of $\theta \in U$). Observe that

$$h(x)\exp(\theta)T(x) \leq \sum_{\pm} \exp(\theta_0 \pm 2\delta)T(x)$$

holds because if $T(x) \geq 0$ then $\exp(\theta T(x))$ is increasing in θ, otherwise it is decreasing and thus the upper bound is obtained. We can thus let $K(x)$ be this bound (it does not depend on $\theta \in U$ but does on θ_0 and thus only on U, which is fine).

Thus we can differentiate under the integral. We can continue in this way (by replacing $h(x)$ with $h(x)T(x)$) and conclude that θ is smooth.

The second claim now follows since $C'(\theta)$ and $C(\theta)^{-1}$ bounded on compact subsets of Θ_0, so the integrand on the right hand side can be uniformly bounded by integrable functions already constructed in the previous arguments:

$$\begin{aligned} |\varphi(x)\,(C'(\theta)h(x)\exp(\theta T(x)) + C(\theta)h(x)T(x)\exp(\theta T(x)))\,| &\leq |C'(\theta)| \cdot |h(x)\exp(\theta T(x))| \\ &\quad + C(\theta)|h(x)T(x)\exp(\theta T(x))| \end{aligned}$$

but each term has already bounded by $K(x)$ in the previous argument. $\square$

We can compute the derivative above as follows

$$\begin{aligned} \frac{d}{d\theta}(E_\varphi(\theta)) &= \int_{\mathcal{X}} \varphi(x)\,(C'(\theta)h(x)\exp(\theta T(x)) + C(\theta)h(x)T(x)\exp(\theta T(x)))\,d\nu(x) \\ &= C'(\theta)C(\theta)^{-1}E_\theta(\varphi) + E_\theta(T\varphi) \end{aligned}$$

Proposition 4.9. If $\varphi : \mathcal{X} \to [0,1]$ is an unbiased test, where $\Theta_0 = \{\theta_0\}$ and Θ is an open subset of $\mathbb{R}$, then

$$C'(\theta_0)C(\theta_0)^{-1}E_{\theta_0}(\varphi) = -E_{\theta_0}(T\varphi).$$

Proof. By definition of unbiased, the point θ_0, is a minimum of $E_\varphi(\theta)$ and thus has zero derivative. $\square$

Appendix A. Review of Probability

Appendix B. Borel spaces

Lemma B.1. Let X be a second-countable metric space. Then there exists an embedding $\iota : X \hookrightarrow Y$ where Y is compact and ι is a homeomorphism onto its image $\iota(X)$.

Proof. Let $u_1, u_2, \ldots$ be a countable dense subset of X. Suppose that $d(x, x') \leq 1$ for all x, x'. Define

$$\iota : X \to [0,1]^{\mathbb{N}}$$

to be

$$\iota(x) = (d(x, u_i))_{i=1}^{\infty}$$

where $Y = [0,1]^{\mathbb{N}}$ has the product topology, thus is a compact metric space. Continuity and injectivity of ι is clear. Now suppose that $x_1, x_2, \ldots \in X$ and $x \in X$ are such that $\iota(x_i) \to \iota(x)$ as $i \to \infty$. Fix $\epsilon > 0$ and choose u_j so that $d(x, u_j) < \epsilon$. Now for all sufficiently large i we have that $|d(x_i, u_j) - d(x, u_j)| < \epsilon$. This implies that $d(x_i, u_j) < 2\epsilon$. Now

$$d(x, x_i) \leq d(x, u_j) + d(u_j, x_i) \leq \epsilon + 2\epsilon = 3\epsilon.$$

Thus ι^{-1} is continuous on $\iota(X)$. □

Lemma B.2. Let X be a complete metric space, with metric d_X, and suppose that $Y \subset X$ is a subset of X such that the subspace topology on Y is induced by a complete metric d_Y on Y. Then Y is a countable intersection of open subsets of X, and is thus a Borel subset of Y.

Proof. We first show that for each $\epsilon > 0$ and $y \in Y$ there exists an open set $V_{y,\epsilon}$ of X such that $y \in V$ and $\mathrm{diam}_{d_X}(V_{y,\epsilon}) \leq \epsilon$ and $\mathrm{diam}_{d_Y}(V_{y,\epsilon} \cap Y) \leq \epsilon$. To see this, observe since d_X and d_Y induce the same topology on Y, there is an open set $\tilde{V}$ of X such that $\tilde{V} \cap Y$ is the ball of radius $\epsilon/2$ around y in the d_Y metric. Hence we can take

$$V_{y,\epsilon} = \tilde{V} \cap B_{d_X}(y, \epsilon/2).$$

Now let $U_n = \bigcup_{y \in Y} V_{y,1/n}$ (Remark: to avoid axiom of choice, just take the union of y and all the different sets of the form $\tilde{V}$). Observe now that $Y \subset U_n$ for all n. We claim that

$$Y = \bigcap_{n=1}^{\infty} U_n.$$

Thus suppose that $x \in U_n$ for all $n \geq 1$. We must show $x \in Y$. We will show that, by induction on n, that there exists a sequence of open subsets $W_1 \supset W_2 \supset \ldots$ of X and $y_n \in Y \cap W_n$ such that $\mathrm{diam}_{d_X}(W_n) \leq \frac{1}{n}, \mathrm{diam}_{d_Y}(W_n \cup Y) \leq \frac{1}{n}$, $x \in W_n$. We can take $W_1 = V_{y_1,1}$ for some $y_1 \in Y$ since $x \in U_1$. Now suppose W_n has been constructed. Thus there exists $\epsilon > 0$ so that $B_{d_X}(x, \epsilon) \subset W_n$ as W_n is open and $x \in W_n$. Now take $N > n$ such that $1/N < \epsilon$. Now since $x \in U_N$ there exists $y_{n+1} \in Y$ and $W_{n+1} \subset X$ open so that $y_{n+1}, x \in W_{n+1}$ and $\mathrm{diam}_{d_X}(W_{n+1}) \leq 1/N$, $\mathrm{diam}_{d_Y}(W_{n+1} \cap Y) \leq 1/N$. Now observe that $W_{n+1} \subset B_{d_X}(x, \epsilon) \subset W_n$ since if $x' \in W_{n+1}$ then

$$d_X(x, x') \leq \mathrm{diam}_{d_X}(W_{n+1}) \leq 1/N < \epsilon.$$

Thus in summary $y_{n+1} \in W_{n+1}$ is non-empty as desired and $W_{n+1} \subset W_n$ and

$$\mathrm{diam}_{d_Y}(W_{n+1} \cap Y) \leq \frac{1}{N} \leq \frac{1}{n+1}.$$

It now follows that $y_n \to x$ in d_X metric and that $y_1, y_2, \dots$ is a Cauchy sequence in Y. Thus by completeness, we have that $y_n \to y \in Y$ for some $y \in Y$. However

$$d_X(x, y) \le d_X(x, y_n) + d_X(y_n, y) \to 0$$

as $n \to \infty$ since $d_Y(y, y_n) \to 0$ and d_X and d_Y induce the same topology. Thus $x = y \in Y$ as required. $\square$

Corollary B.3. Let X be a second countable complete metric space. Then X is homeomorphic to a Borel subset of the compact metric space $[0,1]^{\mathbb{N}}$. In fact, we can take this Borel subset to be a countable intersection of open subsets.

Definition B.4. A *standard Borel space* is a measurable space $(X, \mathcal{B})$ where X is a second-countable complete metric space and $\mathcal{B}$ is the Borel σ-algebra.

Appendix C. Conditional expectation and probability

C.1. **Conditional expectation.**

Theorem C.1. Let $(X, \mathcal{B}, \mu)$ be a probability space and let $\mathcal{A} \subset \mathcal{B}$ be a sub-σ-algebra of $\mathcal{B}$. Then for $f \in L^1(X, \mathcal{B}, \mu)$ there exists a unique $g \in L^1(X, \mathcal{A}, \mu)$ such that

$$\int_A f d\mu = \int_A g d\mu \quad \text{for all } A \in \mathcal{A}.$$

Definition C.2. We say that the $g \in L^1(X, \mathcal{B}, \mu)$ the *conditional expectation of f w.r.t.* $\mathcal{A}$, and denote it by $g = E(f|\mathcal{A})$.

Proof. If $f \in L^2(X, \mathcal{B}, \mu)$ then we define $E(f|\mathcal{A})$ to be the orthogonal projection of f onto the closed subspace $L^2(X, \mathcal{A}, \mu)$. Indeed, for $A \in \mathcal{A}$ we have that $\mathbb{1}_A \in L^2(X, \mathcal{A}, \mu)$ and so

$$\int_A f d\mu = \langle f, \mathbb{1}_A \rangle = \langle E(f|\mathcal{A}), \mathbb{1}_A \rangle = \int_A E(f|\mathcal{A}) d\mu$$

as required. We will now extend $E(\cdot|\mathcal{A})$ to L^1. By density of L^2 in L^1 it is enough to show that $E(\cdot|\mathcal{A})$ is a bounded operator w.r.t. to the L^1 norm on L^2. For $f \in L^2(X, \mathcal{B}, \mu)$ let $\psi \in L^\infty(X, \mathcal{A}, \mu)$ be such that $|E(f|\mathcal{A})| = \psi E(f|\mathcal{A})$ and $\|\psi\|_\infty \le 1$. Thus

$$\|E(f|\mathcal{A})\|_1 = \int_X \psi E(f|\mathcal{A}) d\mu = \langle E(f|\mathcal{A}), \overline{\psi} \rangle = \langle f, \overline{\psi} \rangle = \int_X \psi f d\mu \le \|f\|_1.$$

Thus $E(\cdot|\mathcal{A})$ has L^1 operator norm at most 1 and thus can be extended to $L^1(X, \mathcal{B}, \mu) \to L^1(X, \mathcal{A}, \mu)$.

For uniqueness, suppose that $g_1, g_2 \in L^1(X, \mathcal{A}, \mu)$ are real valued functions such that

$$\int_A g_1 d\mu = \int_A g_2 d\mu \quad \text{for all } A \in \mathcal{A}.$$

Let

$$A_\epsilon = \{x \in X \mid g_1(x) \le g_2(x) - \epsilon\}$$

and observe that $A_\epsilon \in \mathcal{A}$. Thus

$$\int_A g_2 = \int_A g_1 d\mu \le \int_A g_2 d\mu - \epsilon \mu(A)$$

which implies that $\mu(A_\epsilon) = 0$. Thus $g_1 \ge g_2$ almost everywhere. By symmetry, $g_1 = g_2$ almost everywhere. For g_1, g_2 complex valued uniqueness also follows by equating real and imaginary parts. $\square$

Definition C.3. Let $T : X \to Y$ be a measurable map between measure spaces $(X, \mathcal{B}_X)$ and $(Y, \mathcal{B}_Y)$. We define $T^{-1}\mathcal{B}_Y = \{T^{-1}B \mid B \in \mathcal{B}_Y\}$.

Proposition C.4. Suppose that $(X, \mathcal{B}_X, \mu_X)$ is a probability space, $(Y, \mathcal{B}_Y)$ is a measurable space $T : X \to Y$ is measurable. Then any function $f : X \to \mathbb{R}$ that is $T^{-1}\mathcal{B}_Y$ measurable (that is, $f^{-1}B \in T^{-1}\mathcal{B}_Y$ for all Borel $B \subset \mathbb{R}$) can be written as $f = g \circ T$ for some measurable $g : Y \to \mathbb{R}$.

In other words, if we let $\mu_Y = T^*\mu_X$ be the push-forward measure then we have an isometric embedding

$$L^p(Y, \mathcal{B}_Y, \mu_Y) \hookrightarrow L^p(X, \mathcal{B}_X, \mu_X)$$

given by $g \mapsto g \circ T$ whose image is exactly $L^p(\mathcal{X}, T^{-1}\mathcal{B}_Y, \mu_X)$.

Proof. If $f = \mathbb{1}_A$ for some $A \in T^{-1}\mathcal{B}_Y$ then $A = T^{-1}B$ for some $B \in \mathcal{B}_Y$ and so we can take $g = \mathbb{1}_B$. For $f \geq 0$ note that we can write

$$f = \sum_{i=1}^{\infty} c_i \mathbb{1}_{A_i}$$

for some $c_i \geq 0$ and $A_i \in T^{-1}\mathcal{B}_y$ where the limit is pointwise and so $\mathbb{1}_{A_i} = \mathbb{1}_{B_i} \circ T$ for some $B_i \in \mathcal{B}_Y$. Thus $f = g \circ T$ where

$$g(y) = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}(y)$$

converges pointwise as it is monotonic $c_i > 0$. For arbitrary measurable f, note that we can write it as a linear combination of non-negative functions.

Now note that for $g \geq 0$ we have that

$$\int_X g \circ T d\mu_X = \int_Y g d\mu_Y$$

which can be shown as follows. Writing

$$g(y) = \sum_{i=1}^{\infty} c_i \mathbb{1}_{B_i}(y)$$

and using the monotone convergence theorem we have that

$$\int_X g(Tx) d\mu_X(x) = \sum_{i=1}^{\infty} c_i \int_X \mathbb{1}_{B_i}(Tx) d\mu_X(x) = \sum_{i=1}^{\infty} c_i \mu_X(T^{-1}B_i) = \sum_{i=1}^{\infty} c_i \mu_Y(B_i) = \int_Y g(y) d\mu_Y(y).$$

This shows that for $1 \leq p < \infty$ we have that

$$\|g \circ T\|_p^p = \int_X |g|^p(Tx) \mu_X(x) = \int_Y |g|^p(y) d\mu_Y(y) = \|g\|_p^p$$

and so indeed the embedding is an isometry. $\square$

This proposition motivates us to define conditional expectation with respect to a random variable (or factor map).

Definition C.5. Let $(X, \mathcal{B}, \mu)$ be a probability space and suppose that $T : X \to Y$ is measurable where $(Y, \mathcal{B}_Y)$ is a measure space (i.e., a random variable). Then for $f \in L^p(\mathcal{X}, \mathcal{B}, \mu)$ (for $p = 1, 2, \infty$) we define a function on $Y \to \mathbb{C}$ denoted $y \mapsto E(f|T = y)$ that is characterized for μ-almost all x by

$$E(f|\mathcal{T}^{-1}\mathcal{B}_Y)(x) = E(f|T = y) \quad \text{where } y = Tx.$$

In other words, the function $E(f|\mathcal{T}^{-1}\mathcal{B}_Y) \in L^p(\mathcal{X}, T^{-1}\mathcal{B}_Y, \mu)$ is identified with a function in $L^p(Y, \mathcal{B}_Y, T^*\mu)$ using the isometry in the proposition above and this function at $y \in Y$ is denoted $E(f|T = y)$. Note that $E(f|T = y)$ is uniquely defined for $T^*\mu_X$-almost all $y \in Y$.

Note that $E(f|T = y)$ can be characterized by the property that

$$\int_A E(f|T = y) d\mu_Y(y) = \int_{T^{-1}A} f(x) d\mu_X(x) \quad \text{for all } A \in \mathcal{B}_Y.$$

C.2. **Conditional Probability.**

Theorem C.6 (Conditional probability)**.** Let X be a second-countable complete metric space with Borel σ-algebra $\mathcal{B}$ and suppose $\mathcal{A} \subset \mathcal{B}$ is a sub σ-algebra. Let μ be a probability measure on $(X, \mathcal{B})$. Then there exists $X' \in \mathcal{A}$ with $\mu(X') = 1$ and a family of probability measures $\mu_x^{\mathcal{A}}$ for $x \in X'$ such that the following holds.

- For $\mathcal{B}$-measurable functions $f : X \to \mathbb{C}$ such that $\|f\|_\infty < \infty$, we have that the mapping $X' \to \mathbb{C}$ given by
$$x \mapsto \int_X f(u) d\mu_x^{\mathcal{A}}(u)$$
is $\mathcal{A}$-measurable. The same is true if we replace $\|f\|_\infty < \infty$ with $f(x) \in \mathbb{R}_{\geq 0}$ for all $x \in X$.
- For $f : X \to \mathbb{C}$ with $\|f\|_1 < \infty$ we have that[3]
$$E(f|\mathcal{A})(x) = \int_X f(u) d\mu_x^{\mathcal{A}}(u) \quad \text{for almost all } x \in X.$$

Definition C.7. We call such a family of measures $\mu_x^{\mathcal{A}}$ *conditional probabilities of* μ *w.r.t* $\mathcal{A}$.

Proof. Using Corollary B.3 we may assume X is compact. Thus there exists a countable dense $\mathbb{Q}$-vector subspace V of $C(X)$ such that $1 \in V$. Now for each $f \in V$ choose a geniune function (not an equivalence class of functions) $g_f : X \to \mathbb{C}$ such that

$$g_f(x) = E(f|\mathcal{A})(x)$$

and

$$\|g_f\|_\infty \leq \|f\|_\infty.$$

Now for any $\alpha, \beta \in \mathbb{Q}$ and $f_1, f_2 \in V$ we have that

$$g_{\alpha f_1 + \beta f_2}(x) = \alpha g_{f_1}(x) + \beta g_{f_2}(x) \quad \text{for almost all } x \in X.$$

We also have $g_1(x) = 1$ and $g_f(x) \geq 0$ for positive $f \in V$ (i.e, $f \geq 0$) for almost all $x \in X$. Thus we may choose $X' \subset X$ to be a set with $\mu(X') = 1$ on which all of thse properties hold (as there are only countably many equations). Thus for each $x \in X'$ we have a linear function $\Lambda'_x : V \to \mathbb{C}$ given by

$$\Lambda'_x(f) = g_f(x).$$

Note that Λ'_x is uniformly continuous (since $\|g_f\|_\infty \leq \|f\|_\infty$) and thus it extends to a linear functional $\Lambda_x : C(X) = \overline{V} \to \mathbb{C}$ which also satisfies that

$$|\Lambda_x(f)| \leq \|f\|_\infty.$$

We wish to apply the Riesz representation theorem, thus it remains to show that Λ_x is a positive linear functional. Now suppose $f \in C(X)$ is positive. Then there exists $f_i \in V$ such that $f_i \to f$ uniformly. Now

[3]The right hand side may be undefined on a null set of x.

fix a positive integer q and let $\epsilon = 1/q$. Then $\Lambda_x(\epsilon) = \epsilon$ since $g_1 = 1$. For i sufficiently large we have that we have $\|f_i - f\|_\infty < \epsilon$ and so $f_i + \epsilon \geq 0$ as f is positive. As $f_i + \epsilon \in V$ we have that

$$\Lambda_x(f_i) + \epsilon = g_{f_i+\epsilon}(x) \geq 0.$$

Thus

$$\Lambda_x(f) \geq \Lambda_x(f_i) - \Lambda_x(|f_i - f|) \geq \Lambda_x(f_i) - \epsilon > -2\epsilon.$$

This shows that $\Lambda_x(f) \geq 0$ and thust Λ_x is a positive linear continuous functional on $C(X)$ with $\Lambda_x(1) = 1$. Thus there exists a measure μ_x such that

$$\Lambda_x(f) = \int_X f(u)d\mu_x(u) \quad \text{for all } f \in C(X), x \in X'.$$

In particular this means that

$$E(f|\mathcal{A})(x) = \int_X f(u)d\mu_x(u) \quad \text{for all } f \in V.$$

This means that

$$x \mapsto \int_{X'} f(u)d\mu_x(u) \text{ is } \mathcal{A}\text{-measurable on } X' \tag{9}$$

and

$$\int_A \int_{X'} f(u)d\mu_x(u)d\mu(x) = \int_A f(x)d\mu(x) \quad \text{for all } A \in \mathcal{A} \tag{10}$$

holds for all $f \in V$. By density of V in $C(X)$, this also holds for all $f \in C(X)$ and the integrand is indeed $\mathcal{A}$ measurable (it is a pointwise limit of $\mathcal{A}$ measurable functions). Since the indicator function of an open set is a monotonic limit of continuous functions, we have that (9) and (10) holds for $f = \mathbb{1}_U$ for all $U \subset X$ open. It also holds for closed sets by linearity. Finally, observe that it also holds for an intersection $U \cap C$ where U is open and C is closed since such a set is a countable nested intersection of open sets.

Now let

$$\mathcal{R} = \{\cup_{i=1}^n U_i \cap C_i \mid U_1, \ldots, U_n \text{ are open and } C_1, \ldots, C_n \text{ are closed.}\}$$

be the set of all finite unions of sets that can be written as an intersection of an open set and a closed set. We claim that any element in $\mathcal{R}$ can be written as a finite disjoint union of an intersection of an open set with a closed set. Indeed, any such finite union belongs to a finite σ-algebra generated by such sets but any atom of this σ-algebra must take this form. It now follows that (9) and (10) holds for $f = \mathbb{1}_R$ for all $R \in \mathcal{R}$. By the monotone convergence theorem, these also hold on the monotone class generated by $\mathcal{R}$. Thus by the monotone class lemma, it holds on all Borel sets since $\mathcal{R}$ is an algebra. Any measurable $f : X \to \mathbb{R}_{\geq 0}$ is a monotonic limit of simple functions and so it holds for such f too, which completes the proof for such f. The theorem now follows for general $f : X \to \mathbb{C}$ with $\|f\|_1$ as desired by linearity. □

We now show that the conditional probabilities are almost surely unique and provide some characterizations that are practial to use. In the following proposition, we fix $X, \mathcal{A}, \mu$ as as in Theorem C.6.

Proposition C.8. If $\mu_x^{\mathcal{A}}$ and $\nu_x^{\mathcal{A}}$ are two conditional probabilities, for some $x \in X' \in \mathcal{A}$ of full measure, then $\mu_x^{\mathcal{A}} = \nu_x^{\mathcal{A}}$ for μ-almost all $x \in X$.

Proof. We for each $f \in C(X)$ we have that

$$\int_X f(u) d\mu_x^{\mathcal{A}}(u) = E(f|\mathcal{A})(x) = \int_X f(u) d\nu_x^{\mathcal{A}}(u)$$

holds for all $x \in X_f \in \mathcal{A}$ with $\mu(X_f) = 1$. Applying this to a countable dense set of $f \in C(X)$ we can find a single set $X' \in \mathcal{A}$ with $\mu(X'') = 1$ so that this identity holds for all continuous functions $f \in C(X)$. It follows by uniqueness in Riesz-representation theorem that $\mu_x^{\mathcal{A}} = \nu_x^{\mathcal{A}}$ for all $x \in X''$. □

Proposition C.9. Suppose there is a set $X' \in \mathcal{A}$ of full measure and a family of probability measures μ_x for $x \in X'$. Then μ_x form conditional probabilities of μ with respect to $\mathcal{A}$ if and only if the following holds: For all Borel $B \subset X$ we have

$$x \mapsto \mu_x(B)$$

is $\mathcal{A}$ measurable on X' and for all $A \in \mathcal{A}$ we have that

$$\int_A \mu_x(B) d\mu(x) = \mu(A \cap B).$$

Proof. This equation may be written as

$$\int_A \int_X f(u) d\mu_x(u) d\mu(x) = \int_A f(x) d\mu(x)$$

where $f = \mathbb{1}_B$. Thus the μ_x are conditional probabilities if and only if this holds for all continuous f and thus if and only if it holds for all positive functions f. But by the monotone convergence theorem, we have that if it holds for all indicator functions it holds for all positive functions. □

C.3. **Conditional probability with respect to a random variable.** In this subsection, $\mathcal{A}$ is a countable generated σ-algebra of $\mathcal{B}$ where $(X, \mathcal{B})$ is standard Borel. So $\mathcal{A} = \sigma(\{A_1, A_2, \dots, \})$ where $A_1, A_2, \dots \in \mathcal{A}$. For each $x \in X$ we can define the atom

$$[x]_{\mathcal{A}} = \bigcap_{i, x \in A_i} A_i \cap \bigcap_{j, x \notin A_j} (X \setminus A_j)$$

containing x.

Proposition C.10. For $x \in X$ we have that $[x]$ is the smallest subset of $\mathcal{A}$ containing x.

Proof. We must show that for each $A \in \mathcal{A}$ we have either $[x] \subset \mathcal{A}$ or $[x] \cap \mathcal{A} = \emptyset$. This is true for each A_i and so one must show that the sets in $\mathcal{A}$ satisfying this property forms a σ-algebra. □

Proposition C.11. There is a set $X'' \subset X$ with $X'' \in \mathcal{A}$ and $\mu(X'') = 1$ such that for any $x_1, x_2 \in X''$ we have that $\mu_{x_1}^{\mathcal{A}} = \mu_{x_2}^{\mathcal{A}}$ if and only if $[x_1] = [x_2]$. Moreover, $\mu_x^{\mathcal{A}}([x]) = 1$ for $x \in X''$ and if $x' \in X'' \setminus [x]$ then $\mu_x^{\mathcal{A}}([x']) = 0$.

Proof. Let $\mu_x = \mu_x^{\mathcal{A}}$ for $x \in X'$ be a set of conditional probabilities, where $X' \in \mathcal{A}$ and $\mu(X') = 1$. For almost all $x \in X'$ we have that

$$\mu_x(A_i) = \int \mathbb{1}_{A_i} d\mu_x = E(\mathbb{1}_{A_i}|\mathcal{A})(x) = \mathbb{1}_{A_i}(x).$$

Say this occurs on $X'' \subset X'$ with $\mu(X'') = 1$. Note that since all terms are $\mathcal{A}$-measurable, we have $X'' \in \mathcal{A}$. Now let $x_1, x_2 \in X''$. If $\mu_{x_1} = \mu_{x_2}$ then $\mathbb{1}_{A_i}(x_1) = \mathbb{1}_{A_i}(x_2)$ for all i and thus $[x_1] = [x_2]$. Conversely, suppose that $[x_1] = [x_2]$. Thus any $\mathcal{A}$-measurable function must have equal values on x_1 and x_2. But for any $f \in C(X)$ we have that

$$x \mapsto \int f d\mu_x$$

is $\mathcal{A}$-measurable. Thus

$$\int f d\mu_{x_1} = \int f d\mu_{x_2}$$

holds for all $f \in C(X)$ and thus $\mu_{x_1} = \mu_{x_2}$. Now we show the remaining claim. Write

$$[x] = \bigcap_{i \in I} A_i \cap \bigcap_{j \in J} A_j^c.$$

Now observe that for $i \in I$ we have

$$\mu_x(A_i) = \mathbb{1}_{A_i}(x) = 1$$

while for $j \in J$ we have

$$\mu_x(A_j^c) = 1 - \mu_x(A_j) = 1 - \mathbb{1}_{A_j}(x) = 1$$

and so $[x]$ is the intersection of μ_x-co-null sets and this $\mu_x([x]) = 1$. As the atoms form a partition of X'', we have that $\mu_x([x']) = 0$ for $[x'] \neq [x]$. □

Theorem C.12 (Law of total probability)**.** Suppose that $T : X \to Y$ is a measurable map between standard Borel spaces and let $\mathcal{B}_T := T^{-1}\mathcal{B}_Y$ be the induced σ-algebra. There exists Borel $Y' \subset Y$ and probability measures μ_y on X, for each $y \in Y'$, such that the following hold.

- $\mu(T^{-1}Y') = 1$
- For all $y \in Y'$, the measure μ_y is supported on $T^{-1}(\{y\})$.
- For all $x \in T^{-1}(Y')$ we have that $\mu_{Tx} = \mu_x^{\mathcal{B}_T}$. That is the family μ_{Tx} for $x \in T^{-1}(Y')$ is a family of conditional probability measures for the σ-algebra $\mathcal{B}_T$.

Moreover, for bounded Borel $f : X \to \mathbb{C}$ and $B \subset Y$ Borel we have that

$$\int_{T^{-1}B} f(x) d\mu(x) = \int_B \left(\int_X f(u) d\mu_y(u) \right) d(T^*\mu)(y).$$

Proof. The induced σ-algebra $\mathcal{B}_T := T^{-1}\mathcal{B}_Y$ is countably generated and its atoms are precisely the fibers

$$T^{-1}(\{y\}) \quad \text{for } y \in Y.$$

In other words,

$$[x]_{\mathcal{B}_T} = T^{-1}(\{Tx\}) \quad \text{for } x \in X.$$

We thus have by the above proposition that the conditional measure $\mu_x^{\mathcal{B}_T}$ is supported on the fiber $T^{-1}(\{Tx\})$ and only depends on the fiber. More precisely, we have a family of measures μ_y on X, $y \in Y'$ for some Borel $Y' \subset Y$ such that μ_y is supported on $T^{-1}(\{y\})$ and such that for bounded Borel $f : X \to \mathbb{C}$ we have that

$$x \mapsto \int_X f(u) d\mu_{Tx}(u)$$

is $\mathcal{B}_T$ measurable and

$$\int_A f(x) d\mu(x) = \int_A \int f(u) d\mu_{Tx}(u) d\mu(x) \quad \text{for } A \in \mathcal{B}_T.$$

Now be Proposition C.4 we have that there exists $g : Y' \to \mathbb{C}$ which is Borel so that

$$g(Tx) = \int_X f(u) d\mu_{Tx}(u).$$

In particular, this means that

$$y \mapsto g(y) = \int_X f(u) d\mu_y(u)$$

is Borel on Y'.

Now for Borel $B \subset Y'$ we have that $A = T^{-1}B \in \mathcal{B}_T$ and the equation above says that

$$\begin{aligned}\int_A f(x) d\mu(x) &= \int_A g(Tx) d\mu(x) \\ &= \int_B g(y) d(T^*\mu)(y) \\ &= \int_B \left(\int_X f(u) d\mu_y(u) \right) d(T^*\mu)(y)\end{aligned}$$

which completes the proof. □

It is customary to define

$$\mu(B \mid T = y) := \mu_y(B)$$

and

$$E(f|T = y) := E(f|\mathcal{B}_T)(x)$$

for $y = Tx$. The law of total probability then says

$$\mu(A \cap T^{-1}B) = \int_B \mu(A|T = y) d\mu_Y(y)$$

for any Borel $A \subset X, B \subset Y$ where $\mu_Y = T^*\mu$.

Appendix D. Conditional Independence

In this section, fix a standard Borel probability space (Ω, μ) and suppose we have maps $X : \Omega \to \mathcal{X}$, $Y : \Omega \to \mathcal{Y}$ and $T : \Omega \to \mathcal{T}$ to some standard Borel spaces. We also let $\mathcal{A}_X, \mathcal{A}_Y, \mathcal{A}_T$ be the σ-algebras induced by these masp, respectively.

Definition D.1. We say that X and Y are conditionally independent given T if for all $A \subset \mathcal{X}, B \subset \mathcal{Y}$ we have that

$$\mu(X^{-1}A \cap Y^{-1}B|T = t) = \mu(X^{-1}A|T = t)\mu(Y^{-1}B|T = t) \quad \text{for } T^*\mu\text{-almost all } t \in \mathcal{T}.$$

In particular, if T is constant then we say that X and Y are independent, i.e., X and Y are independent if and only if for all $A \subset \mathcal{X}, B \subset \mathcal{Y}$ we have

$$\mu(X^{-1}A \cap Y^{-1}B) = \mu(X^{-1}A)\mu(Y^{-1}B).$$

Example D.2. Suppose $\Omega = [0,1] \times [0,1] \cup [1,2] \times [1,2]$ is the disjoint union of two squares and μ is the uniform probability measure on Ω. Let $X, Y : \Omega \to [0,2]$ denote the projections $X(x,y) = x, Y(x,y) = y$ and let $T : \Omega \to \{0,1\}$ be the map $T(x,y) = \mathbb{1}_{[0,1]^2}((x,y))$. We will show that X, Y are conditionally independent given T but are not independent. First, note that for $U \subset \Omega$ we have that

$$\mu(U) = \frac{1}{2}\lambda_2(U)$$

where λ_d denotes the Lebesgue measure on $\mathbb{R}^d$. It now follows that

$$\mu(U|T = 0) = \lambda_2(U \cap [0,1]^2)$$

and

$$\mu(U|T=1) = \lambda_2(U \cap [1,2]^2).$$

Now for $A, B \subset [0,2]$ we have that

$$\begin{aligned}\mu(X^{-1}A \cap Y^{-1}B|T=0) &= \lambda_2(A \times B \cap [0,1]^2)\\ &= \lambda_1(A \cap [0,1])\lambda_1(B \cap [0,1])\\ &= \lambda_2(X^{-1}A \cap [0,1]^2)\lambda_2(Y^{-1}B \cap [0,1]^2)\\ &= \mu(X^{-1}A|T=0)\mu(Y^{-1}B|T=0)\end{aligned}$$

and a similar calculation holds for conditioning on $T = 1$, which shows conditional independence.

Independence does not hold since if we take $A = [0,1], B = [1,2]$ then

$$\mu(X^{-1}A \cap Y^{-1}B) = \mu([0,1]^2 \cap [1,2]^2) = 0$$

but

$$\mu(X^{-1}A) = \mu([0,1]^2) = \frac{1}{2}$$

and

$$\mu(Y^{-1}B) = \mu([1,2]^2) = \frac{1}{2}.$$

Example D.3 (Independent but not conditionally)**.** Let $\Omega = [0,1]^2$ and suppose μ is the uniform measure on Ω. Let $X, Y : \Omega \to [0,1]$ be the projections. Then X and Y are independent. Let $T : \Omega \to [0,2]$ by $T(x,y) = x + y$. Then in fact X and Y are not conditionally independent given T. For example, let $A = B = [0,1/2]$. Then for $1 < t < \frac{3}{2}$ we have that $(A \times B) \cap T^{-1}(t) = \emptyset$ and so $P(A \times B|T=t) = 0$. However $P(X^{-1}A|T=t) > 0$ and $P(Y^{-1}B|T=t) > 0$.

Observe that we have a map $S : \Omega \to \mathcal{Y} \times \mathcal{T}$ given by $S(\omega) = (Y(\omega), T(\omega))$. We let

$$\mu(A|Y=y, T=t) = \mu(A|S=(y,t).$$

Lemma D.4. We have that X and Y are conditionally independent given T if and only if for all $A \subset \mathcal{X}$ we have that

$$\mu(X^{-1}A|Y=Y\omega, T=T\omega) = \mu(X^{-1}A|T=T\omega) \quad \text{for } \mu\text{-almost all } \omega \in \Omega.$$

Proof. First assume X and Y are conditionally independent. Thus we must show that the right hand side satisfies the definition of conditional probability with respect to the random variable S. The right hand side is S measurable, as required. Thus it remains to show that

$$\int_{S^{-1}U} \mu(X^{-1}A \mid T=T\omega)d\mu(\omega) = \mu(X^{-1}A \cap S^{-1}U)$$

for all $U \subset \mathcal{Y} \times \mathcal{T}$ and $A \subset \mathcal{X}$. By the monotone class lemma, it is enough to show it for $U = B \times C$. We have

$$\begin{aligned}
\int_{S^{-1}(B\times C)} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) &= \int_{Y^{-1}B\cap T^{-1}C} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} \mathbb{1}_{Y^{-1}B}(\omega)\mathbb{1}_{T^{-1}C}(\omega)\mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} E(\mathbb{1}_{Y^{-1}B} \mid T = T\omega)\mathbb{1}_{T^{-1}C}(\omega)\mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{T^{-1}C} \mu(Y^{-1}B|T = T\omega)\mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{T^{-1}C} \mu(Y^{-1}B \cap X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \mu(Y^{-1}B \cap X^{-1}A \cap T^{-1}C) \\
&= \mu(X^{-1}A \cap S^{-1}(B \times C))
\end{aligned}$$

as required.

Now we prove the converse. Thus we must assume that for $A \subset \mathcal{X}$ we have

$$\mu(X^{-1}A|Y = Y\omega, T = T\omega) = \mu(X^{-1}A|T = T\omega) \quad \text{for } \mu\text{-almost all } \omega \in \Omega$$

and show that

$$\int_C \mu(X^{-1}A|T = t)\mu(Y^{-1}B|T = t) dT^*\mu(t) = \int_C \mu(X^{-1}A \cap Y^{-1}B|T = t) dT^*\mu(t)$$

for all $C \subset \mathcal{T}$. Using the law of total probability we have

$$\begin{aligned}
\int_C \mu(X^{-1}A \cap Y^{-1}B|T = t) dT^*\mu(t) &= \mu(X^{-1}A \cap Y^{-1}B \cap T^{-1}C) \\
&= \mu(X^{-1}A \cap S^{-1}(B \times C)) \\
&= \int_{B\times C} \mu(X^{-1}A \mid Y = y, T = t) d(S^*\mu)(y, t) \\
&= \int_{Y^{-1}B\cap T^{-1}C} \mu(X^{-1}A \mid Y = Y\omega, T = T\omega) d\mu(\omega) \\
&= \int_{Y^{-1}B\cap T^{-1}C} \mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} \mathbb{1}_{Y^{-1}B}(\omega)\mathbb{1}_{T^{-1}C}(\omega)\mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} E(\mathbb{1}_{Y^{-1}B}|T = T\omega)\mathbb{1}_{T^{-1}C}(\omega)\mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_{\Omega} \mu(Y^{-1}B \mid T = T\omega)\mathbb{1}_{T^{-1}C}(\omega)\mu(X^{-1}A \mid T = T\omega) d\mu(\omega) \\
&= \int_C \mu(Y^{-1}B \mid T = t)\mu(X^{-1}A \mid T = t) dT^*\mu(t)
\end{aligned}$$

$\square$

Appendix E. Useful results

Lemma E.1. Suppose that ν is a σ-finite positive measure on X. Then there exists a probability measure ν' such that $\nu \ll \nu'$ and $\nu' \ll \nu$ (i.e., $\nu(A) = 0$ if and only if $\nu'(A) = 0$).

Proof. We only need to consider the case $\nu(X) = \infty$. There exists a countable partition $X_1, \dots$ of X such that $0 < \nu(X_i) = d_i < \infty$. Now define

$$\nu'(A) = \sum_{i=1}^{\infty} 2^{-i} d_i^{-1} \nu(A \cap X_i).$$

$\square$

Lemma E.2. Let ν be a σ-finite positive measure on X and let $\mathcal{P}$ be a family of probability measures on X such that $P \ll \nu$ for all $P \in \mathcal{P}$. Then there exists $a_1, a_2, \dots \in [0, 1]$ with $\sum_i a_i = 1$ and $P_1, P_2, \dots \in \mathcal{P}$ such that

$$P \ll \sum_{i=1}^{\infty} a_i P_i \quad \text{for all } P \in \mathcal{P}.$$

Proof. By the previous Lemma, we only need to focus on the case that $\nu(X) = 1$. Let

$$\mathcal{Q} = \{\sum_{i=1}^{\infty} a_i P_i \mid \sum_i a_i = 1, a_i \geq 0, P_i \in \mathcal{P}\}$$

be the set of countable convex combinations of measures in $\mathcal{P}$. Observe that $Q \ll \nu$ for all $Q \in \mathcal{Q}$. For $Q \in \mathcal{Q}$ let $f_Q = dQ/d\nu$ be the density of Q and let

$$S_Q = \{x \in X \mid f_Q(x) > 0\}.$$

We remark that f_Q and S_Q are almost surely well defined. Now let $c = \sup_{Q \in \mathcal{Q}} \nu(S_Q) \leq 1$. Let $Q_1, Q_2, \dots \in \mathcal{Q}$ be such that $\nu(S_Q) \to c$. Now there exists Q_0 such that

$$S_{Q_0} = \cup_{i=1}^{\infty} S_{Q_i}$$

and thus $\nu(S_{Q_0}) = c$. We claim that $P \ll Q_0$ for all $P \in \mathcal{P}$, thus completing the proof. Suppose it were not the case. Then there exists $A \subset X$ such that $P(A) > 0$ but $Q_0(A) = 0$. Now let

$$Q' = \frac{1}{2}\left(Q_0 + P\right).$$

We have that $Q_0 \ll Q'$ and so $S_{Q_0} \subset S_{Q'}$. Since $Q_0(A) = 0$ we have that $\nu(A \cap S_{Q_0}) = 0$. On the other hand

$$0 < P(A) = \int_A f_P d\nu \leq 2 \int_A f_{Q'} d\nu$$

and so this means that there is $A' \subset A$ with $\nu(A') > 0$ and $f_{Q'}(x) > 0$ for $x \in A'$. Thus

$$S_{Q_0} \cup A' \subset S_{Q'}$$

and so

$$\nu(S_{Q'}) \geq \nu(S_{Q_0}) + \nu(A') > c,$$

which is a contradiction. $\square$

Appendix F. Functions between Standard Borel Spaces

We start with a generalization of Proposition C.4 to maps into arbitrary standard Borel spaces rather than just $\mathbb{R}$. Recall that a standard Borel space is a measurable space $(X, \mathcal{B}_X)$ where X is a complete secound countable metric space and $\mathcal{B}_X$ is the Borel σ-algebra on X.

Proposition F.1. Suppose that $(X, \mathcal{B}_X)$ and $(Y, \mathcal{B}_Y)$ are measurable space and let $(Z, \mathcal{B}_Z)$ be a standard Borel space. Let $S : X \to Y$ be measurable and let $\mathcal{B}_S = \{S^{-1}B \mid B \in \mathcal{B}_Y\}$ be the sub-σ-algebra of $\mathcal{B}_Y$ induced by S. Suppose $T : X \to Z$ is $(\mathcal{B}_S, \mathcal{B}_Z)$ measurable. Then there exists a measurable $\psi : Y \to Z$ such that $T = \psi \circ S$.

Proof. We know from Corollary B.3 that there exists a homeomorphism $\iota : Z \to \widetilde{Z}$ where $\widetilde{Z}$ is a Borel subset of $[0,1]^{\mathbb{N}}$. This means that we will now assume that $Z \subset [0,1]^{\mathbb{N}}$. Let $\pi_i : [0,1]^{\mathbb{N}} \to [0,1]$ be the projection map and let $f_i = \pi_i \circ T : X \to [0,1]$. Thus

$$T(x) = (f_1(x), f_2(x), \ldots).$$

Now we can apply Proposition C.4 to $f_i : X \to [0,1]$, which is $\mathcal{B}_S$ measurable as required, to get a measurable $\phi_i : Y \to [0,1]$ such that $f_i = \phi_i \circ S$. Now let $\phi : Y \to [0,1]^{\mathbb{N}}$ be given by

$$\phi(y) = (\phi_1(y), \phi_2(y), \ldots).$$

Now let $\psi : Y \to Z$ be defined by

$$\psi(y) = \begin{cases} \phi(y), & \text{if } y \in \phi^{-1}(Z) \\ z_0, & \text{otherwise} \end{cases}$$

where $z_0 \in Z$ is fixed and arbitrary. Note that ψ is indeed measurable as $\phi^{-1}(Z)$ is measurable. Let us show that ψ now satisfies the desired property. If $x \in X$ then

$$\phi(S(x)) = (\phi_1(Sx), \ldots) = (f_1(Sx), \ldots) = T(x) \in Z$$

and thus $S(x) \in \phi^{-1}(Z)$. This means that $\psi(S(x)) = \phi(S(x)) = T(x)$, as desired. $\square$

Proposition F.2. Let $(X, \mathcal{B}_X), (Y, \mathcal{B}_Y), (Z, \mathcal{B}_Z)$ be standard Borel spaces and suppose that μ is a probability measure on $(X, \mathcal{B}_X)$. Suppose that $S : X \to Y$ and $T : X \to Z$ are Borel measurable and let $\mathcal{B}_S$ and $\mathcal{B}_T$ denote the sub-σ-algebras induced by these maps. Suppose that whenever $T(x) = T(x')$ we have that $S(x) = S(x')$. Then T is μ-almost equal to a map that is $\mathcal{B}_S$-measurable.

Proof. We only have to prove that if $f = \mathbb{1}_A$ for some $A \in \mathcal{B}_T$ then f is μ-almost surely equal to some map that is $\mathcal{B}_S$-measurable. So fix such A. By assumption, we have that A is a union of atoms of $\mathcal{B}_S$, i.e., if we let $[x] = S^{-1}(Sx)$ denote the smallest element of $\mathcal{B}_S$ that contains x (see Propostion C.3) then

$$A = \bigcup_{x \in A} [x].$$

Now we have (by Proposition C.11) that there exists $X' \in \mathcal{B}_S$ with $\mu(X') = 1$ such that if $\mu_x = \mu_x^{\mathcal{B}_S}$ denotes the conditional probability then $\mu_x^{\mathcal{B}_S}([x]) = 1$ for each $x \in X'$. We have (by definition of conditional probability) that the map $\tilde{f} : X' \to \mathbb{R}$ given by

$$\tilde{f}(x) = \int_X \mathbb{1}_A(u) d\mu_x(u)$$

is $\mathcal{B}_S$-measurable. Now observe that for $x \in X'$ we have that if $x \in A$ then $[x] \subset \mathcal{A}$ and so this integral is 1. Otherwise, if $x \in X'$ and $x \notin A$ then $[x] \cap \mathcal{A} = \emptyset$ and so this integral is 0. Thus we have shown that for $x \in X'$ we have that $f(x) = \tilde{f}(x)$ and $\tilde{f}$ is $\mathcal{B}_S$ measurable on $X' \in \mathcal{B}_S$, which completes the proof. □

Appendix G. Characteristic and moment generating functions

In this section, we let μ be a finite signed measure on $\mathbb{R}^d$. By the Hahn-Jordan decomposition, this means that there exists a Borel subset $P \subset \mathbb{R}^d$ such that μ is a positive finite measure on P and $-\mu$ is a positive finite measure on $\mathbb{R}^d \setminus P$. Thus $\mu = \mu^+ - \mu^-$ where μ^+ is μ restricted to P and μ^- is $-\mu$ restricted to $\mathbb{R}^d \setminus P$, so both μ^+ and μ^- are finite positive measures. Note that P is μ-almost unique, thus μ^+ and μ^- are unique.

Example G.1. A typical example is where μ has density w.r.t the Lebesgue measure given by a real valued integrable function, i.e., there is a function $g : \mathbb{R}^d \to \mathbb{R}$ such that

$$\|g\|_1 = \int_{\mathbb{R}^d} |g(x)| d\lambda_d(x) < \infty$$

and $\mu(B) = \int_B g(x) d\lambda_d(x)$ where λ_d denotes the Lebesgue measure on $\mathbb{R}^d$. Here $P = \{x \in \mathbb{R}^d \mid g(x) > 0\}$. Thus μ^+ has density given by $\max(g, 0)$ and μ^- has density given by $\max(-g, 0)$.

Definition G.2. We say that an integral of the form $\int_{\mathbb{R}} f d\mu$ *converges absolutely* if $\int_{\mathbb{R}} |f| d\mu^+ < \infty$ and $\int_{\mathbb{R}^d} |f| d\mu^- < \infty$.

Note that in this case, the value of the integral is defined to be

$$\int_{\mathbb{R}} f d\mu = \int_{\mathbb{R}^d} f d\mu^+ - \int_{\mathbb{R}^d} f d\mu^-.$$

It is easy to see that this integral is linear for such f and the dominated convergence theorem applies (by applying it to μ^+ and μ^-).

Lemma G.3. Let $f : \mathbb{R}^d \to \mathbb{C}$ be a bounded function. Then $\mu \mapsto \int f d\mu$ is a linear functional on the space of finite signed measures.

Proof. Suppose that μ_1, μ_2 are finite signed measures and $\nu = \mu_1 + \mu_2$. The Hahn-Jordan decomposition thus gives

$$\nu^+ - \nu^- = \mu_1^+ - \mu_1^- + \mu_2^+ - \mu_2^-,$$

which gives the following equality of positive measures

$$\nu^+ + \mu_1^- + \mu_2^- = \nu^- + \mu_1^+ + \mu_2^+.$$

Now $\mu \mapsto \int f d\mu$ is additive on the set of positive measures. So we obtain

$$\int f d\nu^+ + \int f d\mu_1^- + \int f d\mu_2^- = \int f d\nu^- + \int f d\mu_1^+ + \int f d\mu_2^+.$$

Rearranging this and using the definition $\int f d\mu$ for finite signed measures, we complete the proof. □

For $\omega \in \mathbb{R}^d$ we define the bounded continous function $e_\omega : \mathbb{R}^d \to \mathbb{C}$ given by

$$e_\omega(x) = \exp(2\pi i \langle \omega, x \rangle).$$

Definition G.4. Define the Characteristic Function of μ to be

$$\widehat{\mu}(\omega) = \int_{\mathbb{R}^d} e_\omega(x) d\mu(x).$$

The characteristic function is also known as the Fourier transform.

Theorem G.5. Suppose that

$$\widehat{\mu}(\omega) = 0 \quad \text{for all } \omega \in \mathbb{R}^d.$$

Then $\mu = 0$ is the zero measure on $\mathbb{R}^d$.

Proof. This follows from Fourier Inversion for Tempered Distributions, but let us give a simpler proof using just the more elementary Stone-Weierstrass Theorem. First, choose a constant M so that $\mu^+(B) + \mu^-(B) < M$ for all $B \subset \mathbb{R}^d$. It will be enough to show that if $\phi : \mathbb{R}^d \to \mathbb{R}$ is a compactly supported continuous function, then

$$\int_{\mathbb{R}^d} \phi(x) d\mu(x) = 0.$$

Now take R large enough so that the support of ϕ is contained inside the box $(-R/2, R/2)^d$. Now fix $\epsilon > 0$. By the Stone-Weierstrass theorem, we have a trigonometric polynomial

$$P_R = \sum_{i=1}^{k} c_{i,R} e_{\omega_{i,R}}$$

such that

$$|f(x) - P_R(x)| < \epsilon \quad \text{for all } x \in [-R/2, R/2]^d$$

and P_R is R-periodic (by this we mean that $P_R(x+v) = P_R(x)$ for all $v \in R\mathbb{Z}^d$). Thus if we let $\phi_R : \mathbb{R}^d \to \mathbb{R}$ be a function that equals to ϕ_R on the box $(-R/2, R/2)^d$ but is R-periodic on $\mathbb{R}^d$ then we have that

$$\|\phi_R - P_R\|_\infty < \epsilon.$$

It follows now that

$$\left| \int_{\mathbb{R}^d} \phi_R d\mu - \int_{\mathbb{R}^d} P_R d\mu \right| < M\epsilon.$$

However

$$\int_{\mathbb{R}^d} P_R d\mu = \sum_{i=1}^{k} c_{i,R} \hat{\mu}(\omega_{i,R}) = 0.$$

Thus we have that

$$\left| \int_{\mathbb{R}^d} \phi_R d\mu \right| < M\epsilon.$$

However note that

$$\phi_R \to \phi \quad \text{as R } \to \infty$$

pointwise and $|\phi_R(x)| \leq \|\phi\|_\infty$ for all $x \in \mathbb{R}^d$, thus the dominated convergence theorem applies. This shows that

$$\left| \int_{\mathbb{R}^d} \phi d\mu \right| < M\epsilon.$$

This completes the proof, as ϵ is arbitrary and M is constant. ☐

Corollary G.6. If two finite signed measures have the same characteristic functions, then they are equal.

Proof. By G.3 the map $\mu \mapsto M_\mu$ is a linear map (from the space of signed measures to the space of complex functions on $\mathbb{R}^d$). By the previous theorem, it has a trivial kernel so is injective. ☐

Note that if $T : X \to Y$ is a measurable map and μ is a finite signed measure on X, then the push-forward is defined as usual: $T^*\mu(B) = \mu(T^{-1}B)$ for $B \subset X$. Thus $\mu_T := T^*\mu$ is a positive finite signed measure.

Example G.7 (Warning!!)**.** Note that $T^*\mu = T^*\mu^+ - T^*\mu^-$ but $T^*\mu^+$ may not equal $(T^*\mu)^+$. For example, if $T : \{0,1\} \to \{0\}$ is the constant map, and $\mu(\{0\}) = -1$, $\mu(\{1\}) = 1$ then $T * \mu = 0$ is the zero measure and so $(T^*\mu)^+$ is the zero measure, yet $T^*\mu^+$ is the probability meausre on $\{0\}$.

Note that for bounded $f : Y \to \mathbb{C}$ we still have that $\int f dT^*\mu = \int (f \circ T) d\mu$ holds. To see this, note that

$$\int_Y f dT^*\mu = \int_Y f dT^*(\mu^+ - \mu^-) = \int_Y f d(T^*\mu^+ - T^*\mu^-) = \int_Y f d(T^*\mu^+) - \int_Y f d(T^*\mu^-),$$

where in the last equation we used Lemma G.3.

Corollary G.8. Let $L_v : \mathbb{R}^d \to \mathbb{R}$ be the projection $L_v(x) = \langle v, x\rangle$. Suppose that μ and ν are two finite signed measures on $\mathbb{R}^d$ such that for all $v \in \mathbb{R}^d$ we have that $L_v^*\mu = L_v^*\nu$. Then $\mu = \nu$.

Proof. We have that

$$\widehat{\mu}(v) = \int_{\mathbb{R}^d} \exp(2\pi i\langle x, v\rangle) d\mu(x) = \int_{\mathbb{R}} \exp(2\pi i t)(L_v^*\mu)(t) = \int_{\mathbb{R}} \exp(2\pi i t)(L_v^*\nu)(t) = \widehat{\nu}(v),$$

thus the characerteristic functions are the same. □

To define the moment generating function of μ, we would like to extend the characteristic function to complex values. The issue is that we may have non-converging integrals.

Define the moment generating function to be

$$M_\mu(z) = \int_{\mathbb{R}} \exp(xz) d\mu(x)$$

whose domain is the set of all $z \in \mathbb{C}$ where this integral converges absolutely (in the sense of Definition G.2).

Proposition G.9. Suppose that $U \subset \mathbb{C}$ is an open set containing the domain of M_μ (i.e., the integral converges absolutely for $z \in U$). Then M_μ is holomorphic on U and its derivative on U is given by

$$M'_\mu(z) = \int_{\mathbb{R}} x \exp(xz) d\mu(x),$$

and this integral converges absolutely for $z \in U$.

Proof. Fix $z \in U$. We have

$$\frac{M_\mu(z + w) - M_\mu(z)}{w} = \int_{\mathbb{R}} \frac{e^{x(z+w)} - e^{xz}}{w} d\mu(x)$$

for small enough $w \in \mathbb{C}$ (as U is open). Thus, by the Lebesgue dominated convergence theorem, it will be enough to show that for small w, the integrand is bounded by a single μ-integrable function (i.e., a function integrable w.r.t both μ^+ and μ^-). We can now use the general mean-value inequality

$$|f(z + w) - f(z)| \le |w| \sup_{t\in[0,1]} |f'(z + tw)|$$

to $f(z) = e^{xz}$ to get that

$$\left| \frac{e^{x(z+w)} - e^{xz}}{w} \right| \le \sup_{t\in[0,1]} |x \exp(x(z + wt))| = |x|| \exp(xz)| \sup_{t\in[0,1]} |\exp(xwt)|.$$

Now as x is real, the maximum occurs at either $t = 0$ or $t = 1$ depending on whether xw has positive or negative real part. Thus

$$\left|\frac{e^{x(z+w)} - e^{xz}}{w}\right| \leq |x||\exp(xz)|(|\exp(xw)| + 1)$$

Now let ϵ be so small that the ball of radius 2ϵ around z is inside u. We will now assume $|w| < \epsilon$. Then there is a constant $C_\epsilon > 0$ so that

$$|x| < C_\epsilon(e^{\epsilon x} + e^{-\epsilon x}) \quad \text{for all } x \in \mathbb{R}.$$

Thus we get that

$$\left|\frac{e^{x(z+w)} - e^{xz}}{w}\right| \leq C_\epsilon(e^{\epsilon x} + e^{-\epsilon x})|\exp(xz)|(|\exp(xw)| + 1).$$

Thus after expanding these factors in the upper bound, we get a sum of terms of the form

$$C|\exp(x(z + \delta))|$$

for some $\delta \in \mathbb{C}$ with $|\delta| < 2\epsilon$ and thus $z + \delta \in U$ so such a term is integrable w.r.t to μ, by assumption. Thus the dominated convergence theorem applies. $\square$

Corollary G.10. Suppose that there exists $\epsilon > 0$ so that for $t \in (-\epsilon, \epsilon)$ we have that

$$\int_{\mathbb{R}} |\exp(tx)| d\mu(x) < \infty.$$

If $M_\mu(t) = 0$ for all $t \in (-\epsilon, \epsilon)$ then $\mu = 0$ is the zero measure. In other words, if two finite signed measures have the same (well-defined) characteristic function on some interval containing 0, then they must be equal.

Proof. Let $U = (-\epsilon, \epsilon) \times \mathbb{R} \subset \mathbb{C}$ and note that for $z = t + \omega i \in U$, where $\omega \in \mathbb{R}$ and $t \in (-\epsilon, \epsilon)$ we have that

$$|\exp(xz)| = |\exp(xt)|$$

and so $M_\mu(z)$ is indeed defined (the definining integral converges absolutely) for $z \in U$. It is holomorphic on U and thus, as it vanishes on the interval $(-\epsilon, \epsilon)$ and it vanishes on all of U. In particular, it vanishes on the imaginary line $\mathbb{R}i$ and so the characteristic function of μ vanishes. Thus $\mu = 0$. $\square$

We can define the moment generating functions for finite signed measures μ on $\mathbb{R}^d$ as follows. If $v \in \mathbb{R}^d$ then

$$M_\mu(v) = \int_{\mathbb{R}^d} \exp(\langle v, x\rangle) d\mu(x)$$

where the domain of M_μ is the set of those v where the integral converges absolutely.

Theorem G.11. Suppose that the domain of M_μ contains an open subset $U \subset \mathbb{R}^d$ such that $0 \in U$. Suppose that $M_\mu(u) = 0$ for all $u \in U$. Then $\mu = 0$ is the zero measure on $\mathbb{R}^d$.

Proof. From Corollary G.8, it will be enough to show that for all $v \in \mathbb{R}^d$, the pushforward measure $L_v^*(\mu)$ is the zero measure on $\mathbb{R}^d$. Now fixing $v \in \mathbb{R}^d$, there exists an ϵ such that $tv \in U$, for all $t \in (-\epsilon, \epsilon)$. This means that the following integrals converge absolutely:

$$\begin{aligned} M_\mu(tv) &= \int_{\mathbb{R}^d} \exp(t\langle v, x\rangle) d\mu(x) \\ &= \int_{\mathbb{R}^d} \exp(tL_v(x)) d\mu(x) \\ &= \int_{\mathbb{R}^d} \exp(tL_v(x)) d\mu^+(x) - \int_{\mathbb{R}^d} \exp(tL_v(x)) d\mu^-(x) \\ &= \int_{\mathbb{R}} \exp(ty) d(L_v^*\mu^+)(y) - \int_{\mathbb{R}} \exp(ty) d(L_v^*\mu^-)(y) \end{aligned}$$

But as $tv \in U$ we have that $M_\mu(tv) = 0$ and thus these two integrals are equal. Thus the moment generating function of $L_v^*\mu^+$ and $L_v^*\mu^-$ are equal for $t \in (-\epsilon, \epsilon)$. This means that these two measures are equal and thus L_v^* is the zero measure, as desired. □